ITDS231
Lecture 6.2

Data Link Layer
Ethernet

Asst. Prof. Dr. Thitinan Tantidham

Some slides from
- ITCS323/2022s2
- NETACAD Network Essentials/ITN7.02

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

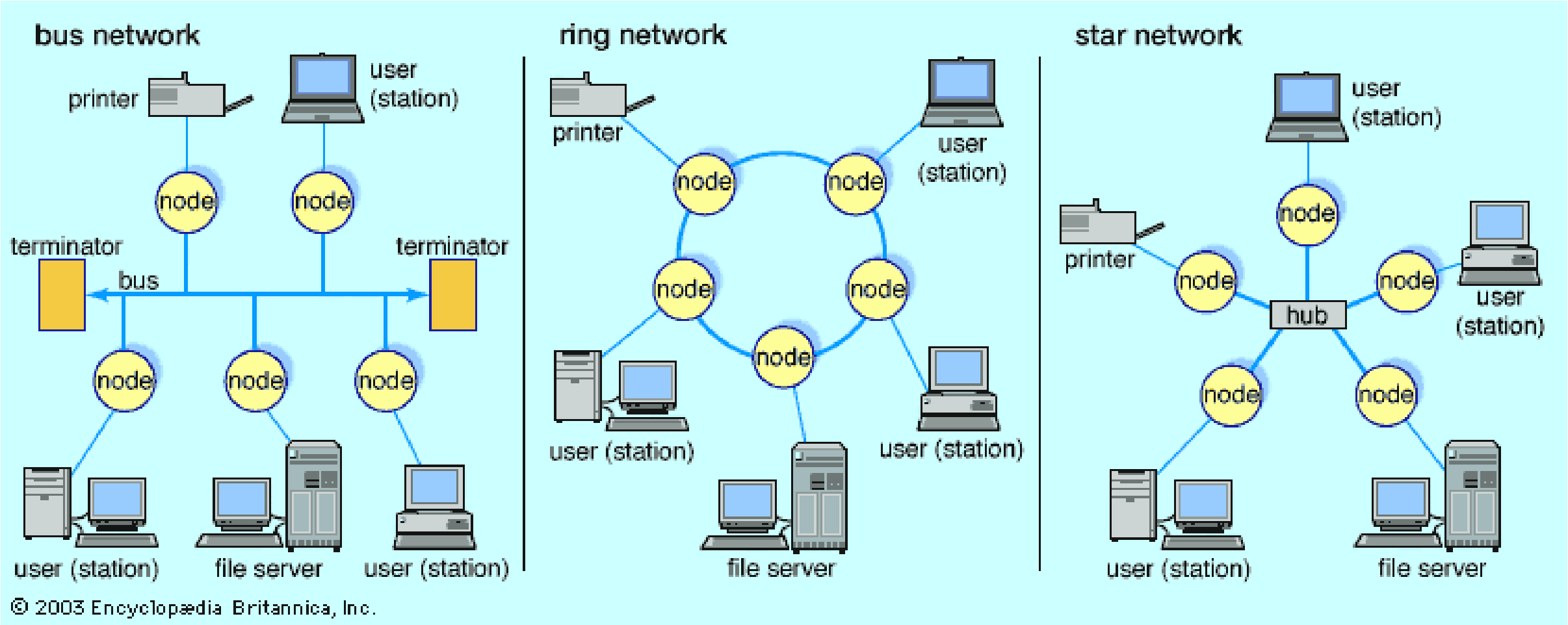
Unless allowed by Mahidol University, the User shall not copy, modify, or exploit in part or in whole of the copyrighted materials on the Platform, other than for your own individual study. The copy, modify or communication to public of the copyrighted materials shall be inflicted with a fine from 20,000 Baht up to 200,000 Baht. If the offence is committed with the commercial purpose, the offender shall be inflicted with imprisonment for a term from 6 months up to 10 years or a fine from 100,000 Baht up to 800,000 Baht or both imprisonment and fine.

Ethernet

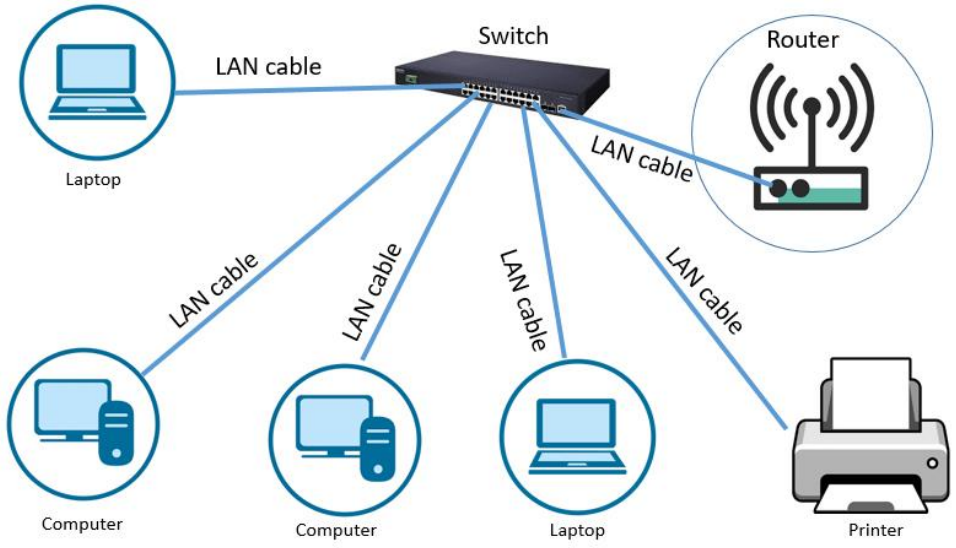
Layer	Type	OSI Model	TCP/IP Model	Authority
7	Data	Application Layer	Application Layer	RFCs, IETF, Industry Orgs, Etc.
7	Data	Presentation Layer		
5	Data	Session Layer		
4	Segments	Transport Layer	TCP/UDP	
3	Packets	Network Layer	IP	
2	Frames	Data Link Layer	Ethernet	IEEE 802.1
1	Bits	Physical Layer		IEEE 802.3
		Network		

- We learn that there are many protocols and techniques in the data-link layer and physical layer. It is usually used in Local Area Network (LAN).
- Too many protocols create too many problems and confusion to the end consumer (too flexible like a PC computer).
- To simplify the protocols, a specific standard (de facto standard) is established. This is called '**Ethernet**' that is easy to use by most people and support the Internet.
- Ethernet is a protocol that covers both data-link layer and physical layer together into one standard.

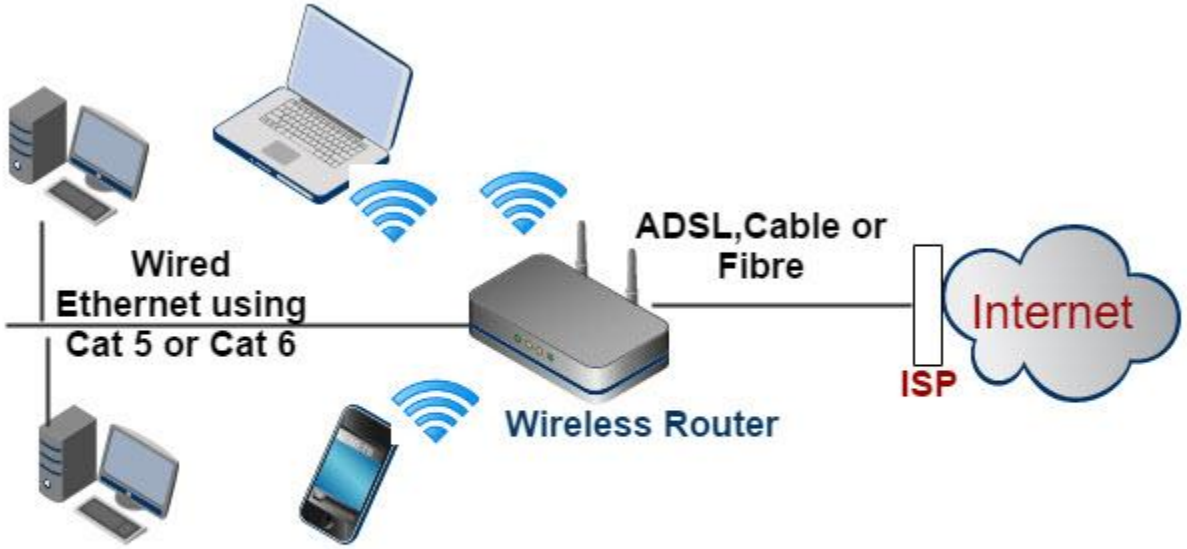
History


bus network
printer
user (station)
node
node
terminator
bus
terminator
node
node
node
user (station)
file server
user (station)
ring network
printer
node
node
user (station)
node
node
node
user (station)
user (station)
file server
star network
user (station)
printer
node
node
node
hub
user (station)
node
node
user (station)
file server
© 2003 Encyclopædia Britannica, Inc.

Today



Local Area Network



Network Diagram-Typical Simple Home Network

History

- Ethernet was developed at Xerox PARC between 1973 and 1974.
- It was inspired by ALOHAnet, which a Xerox employee, Robert Metcalfe, had studied as part of his PhD dissertation.
- Metcalfe left Xerox in 1978 to form 3Com a famous network company (now acquired by HP).

- Matcalfe convinced Digital Equipment Corporation, Intel and Xerox to promote Ethernet as a standard.

- Ethernet initially competed with expensive IBM's Token Ring and other proprietary protocols.
- Ethernet was able to adapt to market needs and with 10BASE2, shift to inexpensive thin coaxial cable and from 1990, to the now-ubiquitous twisted pair with 10BASE-T (UTP) which completely kills Token Ring and other proprietary protocols for the PC market (1995 onwards).

Standard (1/2)

- In February 1980, the Institute of Electrical and Electronics Engineers (IEEE) started project 802 (Ethernet) to standardize local area networks (LAN).
  - Data Link Layer
    - ✓ Framing
    - ✓ Flow control (Ethernet has no flow control)
    - ✓ Error control (CRC-32)
    - ✓ Media Access Control (CSMA/CD for wired 1983 and CSMA/CA for wireless in 1997)
  - Physical layer
    - ✓ Wiring and signaling protocols

- At the same time, Token Ring (supported by IBM), Token Bus (selected and henceforward supported by General Motors) and FDDI (Fiber Distributed Data Interface supported by ANSI) were also considered as candidates for a LAN standard.
  - ✓ Token-based systems were not simple. They need coordination of machines.
  - ✓ Plus, the proposed token-based systems were governed by a specific company. It is not open to the community.

Standard (2/2)

- In 1983 Ethernet becomes IEEE 802.3 standard for Local Area Network used by billions of people around the world.
- Almost every LAN except Ethernet has disappeared from the marketplace because
  - Ethernet was able to evolve to meet the needs of higher transmission rates.
  - Ethernet is open to the community. Anyone can manufacture Ethernet devices as long as it follows IEEE 802.3 specification.
  - Ethernet is a cheaper technology than the other competing technologies.

802.3cu	2021-02	100 Gbit/s and 400 Gbit/s over SMF using 100 Gbit/s lanes
802.3cv	2021-05	Power over Ethernet maintenance, chaired by Chad Jones[8]
802.3cw	(TBD)	400 Gbit/s over DWDM Systems – scheduled for fall 2024, chaired by John D'Ambrosia[8]
802.3cx	2023-03	Improved PTP timestamping accuracy, chaired by Steve Gorshe[8]
802.3cy	2023-06	MultiGigBASE-T1 25 Gbit/s electrical automotive Ethernet,[9] chaired by Steve Carlson[8]
802.3cz	2023-03	Multi-gigabit optical automotive Ethernet, chaired by Bob Grow[8]
802.3da	(TBD)	10BASE-T1S 10 Mb/s operation over single balanced pair multidrop segments, extended length up to 50 m – scheduled for mid 2025, chaired by Chad Jones[8]
802.3db	2022-09	100 Gbit/s, 200 Gbit/s, and 400 Gbit/s operation over optical fiber using 100 Gbit/s Signaling, chaired by Robert Lingle[8]
802.3-2022	2022-07	802.3dc – 802.3-2018 maintenance, merge recent amendments bt/ca/cb/cd/cg/ch/cm/cn/cp/cq/cr/ct/cu/cv, chaired by Adam Healey[8]
802.3dd	2022-06	Power over Data Lines of single-pair Ethernet maintenance, chaired by George Zimmerman[8]
802.3de	2022-09	Time synchronization for point-to-point single-pair Ethernet, chaired by George Zimmerman
802.3df	2024-02	200 Gb/s, 400 Gb/s and 800 Gb/s using 100 Gbit/s lanes, chaired by John D'Ambrosia
802.3dg	(TBD)	100BASE-T1L (100 Mbps single pair with extended length to 500 m) – scheduled for mid 2025, chaired by George Zimmerman
802.3dh	canceled	Multi-gigabit automotive Ethernet over plastic optical fiber, chaired by Yuji Watanabe
802.3dj	(TBD)	200 Gb/s, 400 Gb/s, 800 Gb/s and 1.6 Tbit/s using 200 Gbit/s lanes – scheduled for spring 2026, chaired by John D'Ambrosia
802.3dk	(TBD)	Greater than 50 Gbit/s bidirectional optical access, chaired by Yuanqiu Luo
802.3dm	(TBD)	Asymmetrical Electrical Automotive Ethernet, chaired by Jon Lewis
802.3dn	(TBD)	802.3-2022/Cor 1 Multi-Gigabit Automotive MDI Return Loss, chaired by Brett McClellan

https://en.wikipedia.org/wiki/IEEE_802.3

https://ieeexplore-ieee-org.ejournal.mahidol.ac.th/stamp/stamp.jsp?tp=&arnumber=9844436

IEEE Project 802 (1/2)

- Early Ethernet focused on the standardization of media access control (MAC) operation and CSMA/CD was pushed to be the standard.

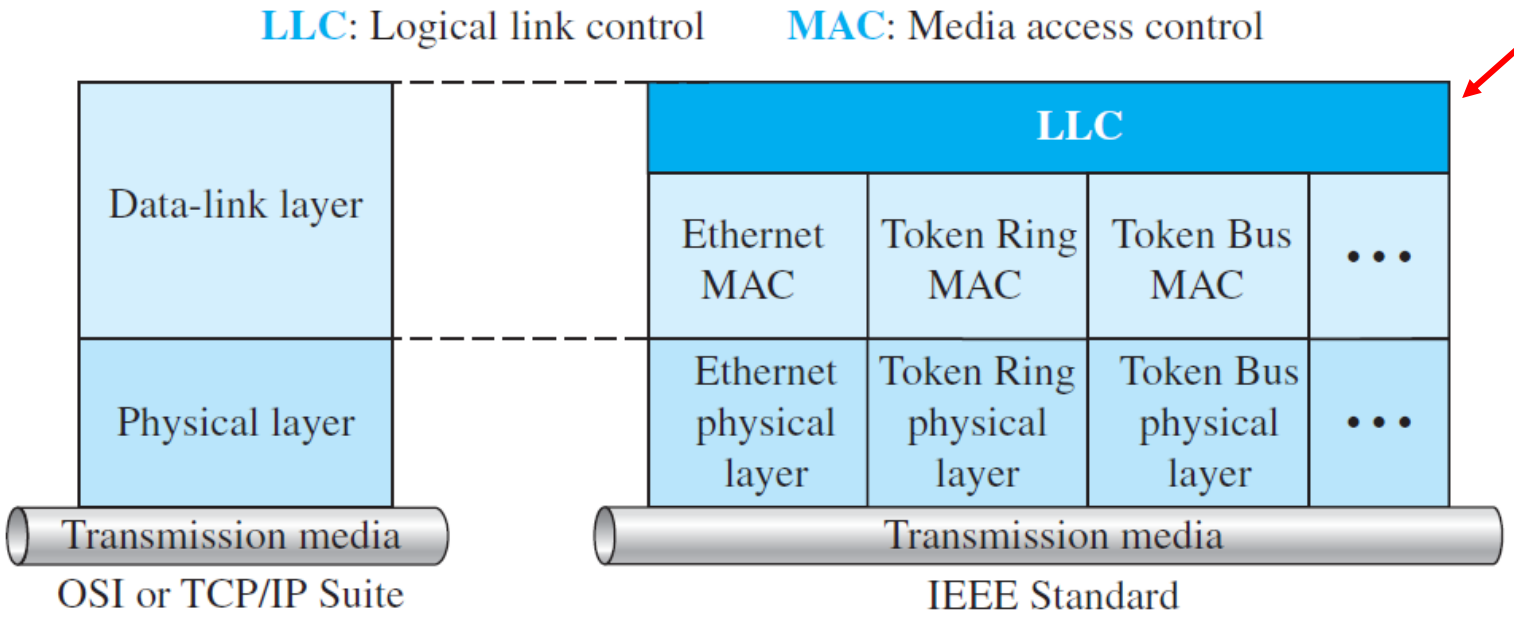
- Today, Ethernet has evolved to include higher bandwidth, improved medium access control methods, and different physical media. The coaxial cable was replaced with point-to-point links connected by Ethernet repeaters or switches.

IEEE Project 802 (2/2)

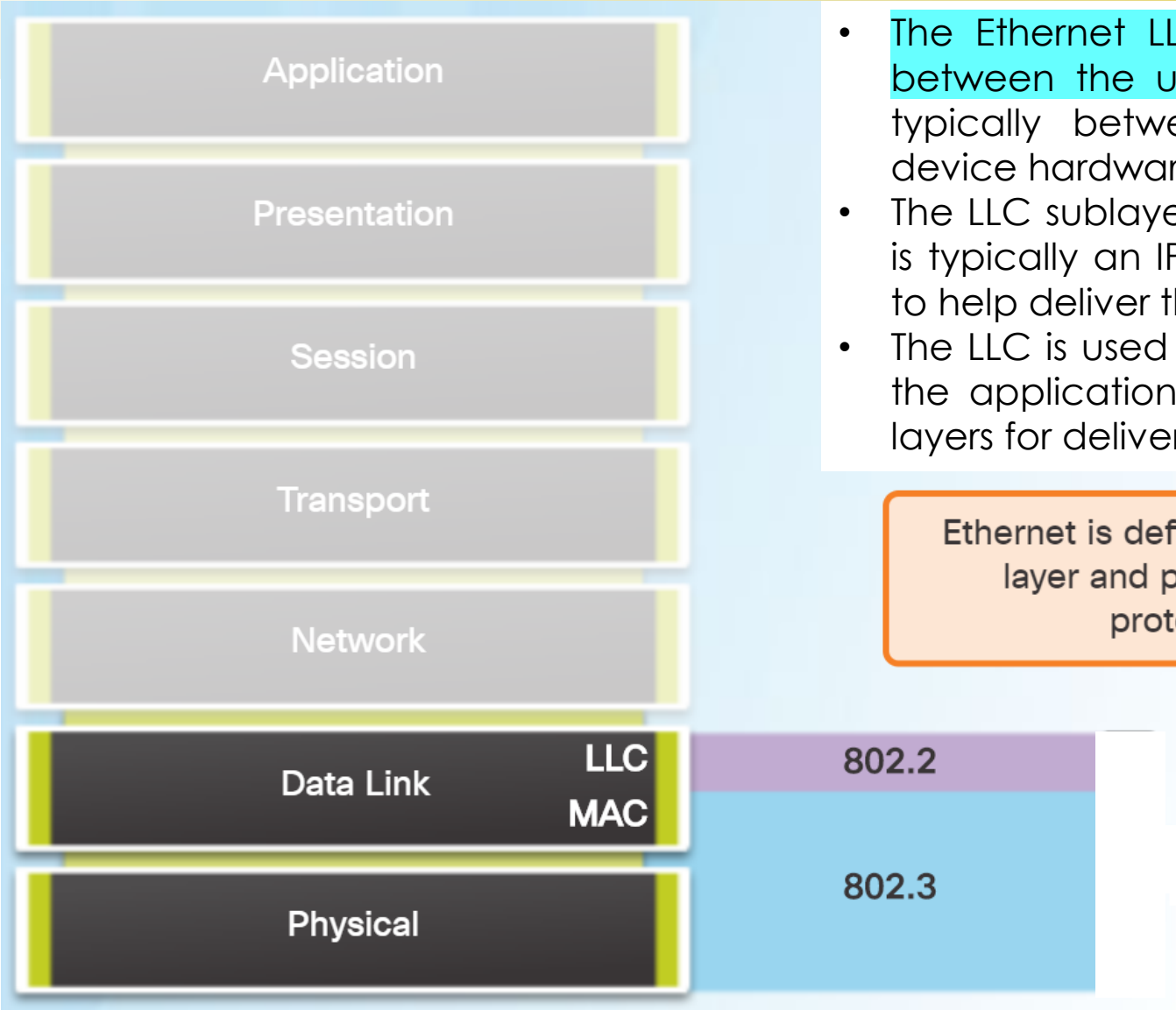
- Project 802 was started to unify functions and specifications of the physical layer and the data-link layer of major LAN protocols because
  - There were too many protocols at the time
  - Physical layer and data-link layer have to work closely together.

- At the time, there were many competing technologies

LLC is not a part of Ethernet



Ethernet in OSI Model



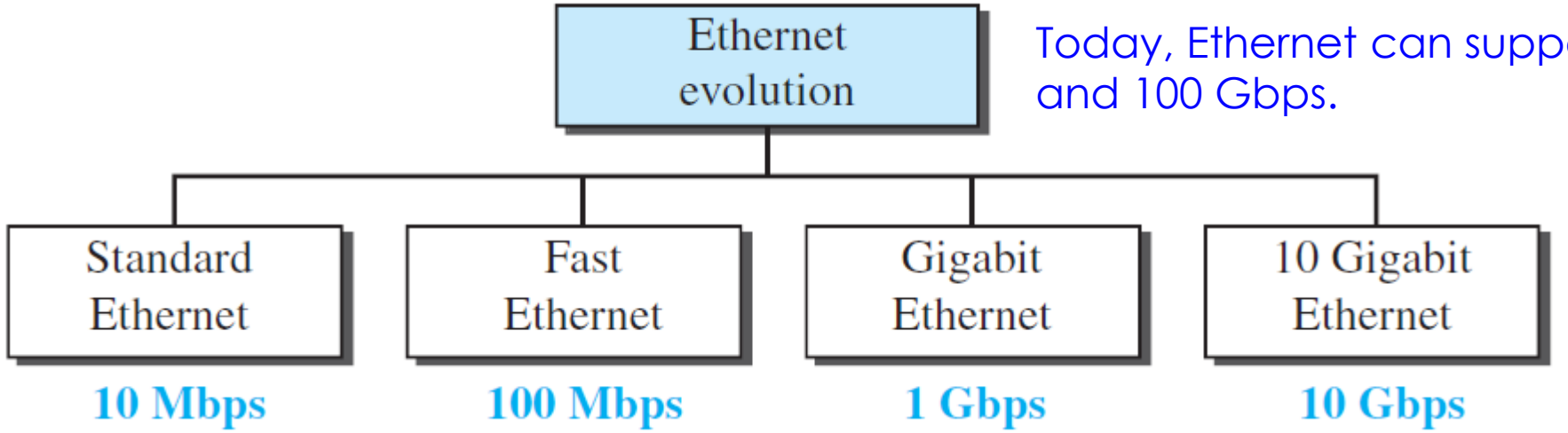
- The Ethernet LLC sublayer handles the communication between the upper layers and the lower layers. This is typically between the networking software and the device hardware.
- The LLC sublayer takes the network protocol data, which is typically an IPv4 packet, and adds control information to help deliver the packet to the destination node.
- The LLC is used to communicate with the upper layers of the application, and transition the packet to the lower layers for delivery.

Ethernet is defined by data link layer and physical layer protocols.

Ethernet == IEEE 802.3

Evolution

- There are 4 major types of Ethernet



Today, Ethernet can support 40 Gbps and 100 Gbps.

Use CSMA/CD (Bus or Hub)

CSMA/CD is optional (hub only)

No CSMA/CD (switch is used)

Flow Control

- Ethernet provides a connectionless service.
  - Each frame sent is independent of the previous or next frame.
  - Ethernet has no connection establishment or connection termination phases.

- Since Ethernet has no LLC, the sender sends a frame whenever it has it. The receiver may or may not be ready for it. The sender may overwhelm the receiver with frames, which may result in dropping frames. If a frame drops, the sender will not know about it.

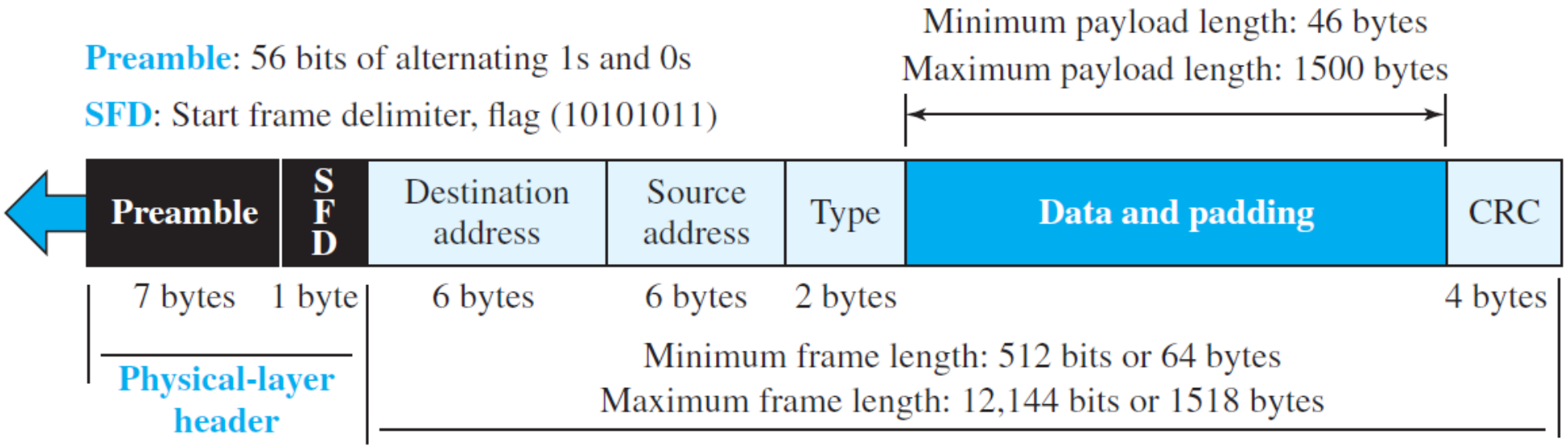
- We rely on the higher level protocol such as TCP to deal with the frame drop using flow control in the transport layer.
  - **Internet protocol** (layer 3), which is using the service of Ethernet, is connectionless, it will not know about a frame drop.
  - If the transport layer is also a connectionless protocol, such as **UDP**, the frame is lost and salvation may only come from the application layer. However, if the transport layer is **TCP**, the sender TCP does not receive acknowledgment for its segment and sends it again.

Error Control

- Ethernet has no LLC (flow control), Ethernet is unreliable like IP and UDP.
- Data link layer is responsible for flow control. Most real-world implementations just use Ethernet without flow control and leave flow control to transport layer for speed.
- Ethernet does provide CRC-32 for frame error checking.
- If a frame is corrupted during transmission and the receiver finds out about the corruption, due to CRC-32, the receiver drops the frame silently.
- It is the duty of high-level protocols to find out about it and correct it using transport layer protocols.

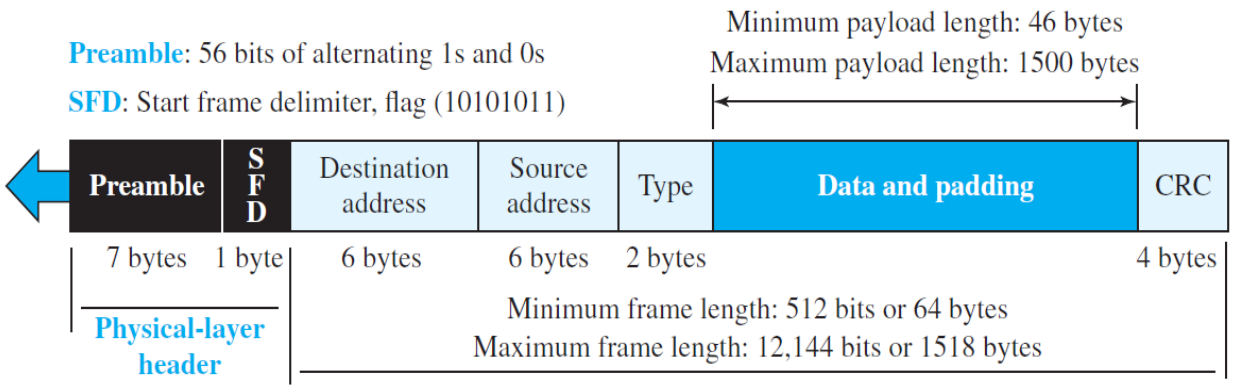
Framing (1/2)

- **Preamble**: This field contains 7 bytes (56 bits) of alternating 0s and 1s that alert the receiving system to the coming frame and enable it to synchronize its clock if it's out of **synchronization**. The preamble is actually added at the physical layer and is not (formally) part of the frame.
- ***Start frame delimiter (SFD):*** This field (1 byte: 10101011) signals the beginning of the frame. The SFD warns the station or stations that this is the last chance for synchronization. This field is actually a flag that defines the beginning of the frame. SFD is added at the physical layer.



Framing (2/2)

- ***Destination address (DA).*** This field is six bytes (48 bits) and contains the link-layer address of the destination station or stations to receive the packet. → *MAC address*
- ***Source address (SA).*** This field is also six bytes and contains the link-layer address of the sender of the packet.
- ***Type.*** This field defines the upper-layer protocol whose packet is encapsulated in the frame. This protocol can be IP, ARP, OSPF, and so on.
- ***Data.*** This field carries data encapsulated from the upper-layer protocols. It is a minimum of 46 and a maximum of 1500 bytes. It has to have a minimum length because of CSMA/CD requirement to listen to the incoming frame. 0-padding is needed if the frame data is too short.
- ***CRC.*** The last field contains error detection information, in this case a CRC-32. The CRC is calculated over the addresses, types, and data field.



Frame Length

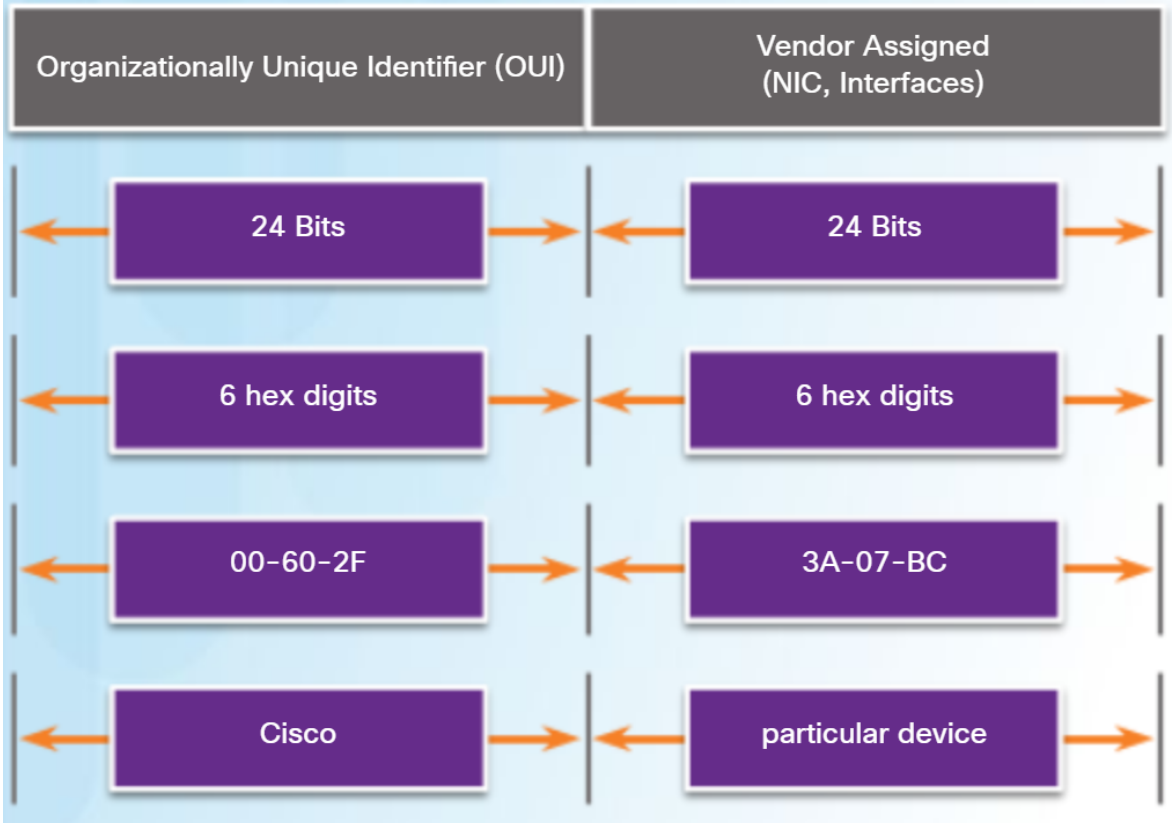
- An Ethernet frame needs to have a minimum length of 512 bits or 64 bytes to support CSMA/CD. Part of this length is the header and the trailer. If we count 18 bytes of header and trailer (6 bytes of source address, 6 bytes of destination address, 2 bytes of length or type, and 4 bytes of CRC), then the minimum length of data from the upper layer is 64 – 18 = 46 bytes. (64 - See this lecture P51)

- The standard defines the maximum length of a frame (*without preamble and SFD field)* as 1518 bytes. If we subtract the 18 bytes of header and trailer, the maximum length of the payload is 1500 bytes. This is because
  - Memory was very expensive when Ethernet was designed; a maximum length restriction helped to reduce the size of the buffer.
  - Second, the maximum length restriction prevents one station from monopolizing the shared medium, blocking other stations that have data to send.

Minimum frame length: 64 bytes	**Minimum data length: 46 bytes**
Maximum frame length: 1518 bytes	**Maximum data length: 1500 bytes**

Addressing

- Each station on an Ethernet network (such as a PC, workstation, or printer) has its own **network interface card (NIC).** The NIC fits inside the station and provides the station with a link-layer address. The Ethernet address is 6 bytes (48 bits), normally written in **hexadecimal notation,** with a colon between the bytes (e.g., **4A:30:10:21:10:1A**). The first 24 bits are manufacturer specific assigned by IEEE (OUI) and the last 24 bits are assigned by manufacturers (vendors).



Note: It is possible for duplicate MAC addresses to exist due to mistakes during manufacturing or in some virtual machine implementation methods.

In either case, it will be necessary to modify the MAC address with a new NIC or in software.

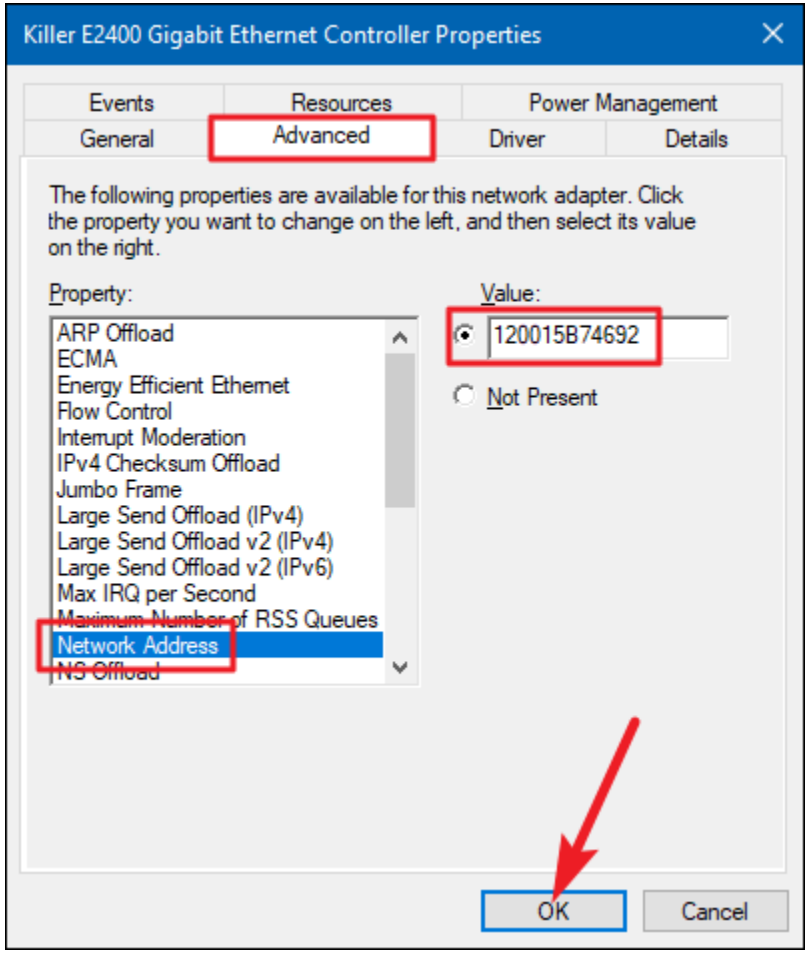
MAC Spoofing

- MAC Address spoofing or MAC spoofing is a technique for changing a factory-assigned Media Access Control (MAC) address of a network interface on a networked device. The MAC address that is hard-coded on a network interface controller (NIC) cannot be changed. However, many drivers allow the MAC address to be changed.
  - MAC address spoofing is limited to the local broadcast domain. Unlike IP address spoofing, where senders spoof their IP address in order to cause the receiver to send the response elsewhere
  - In MAC address spoofing, the response is usually received by the spoofing party if switch is not configured to prevent MAC spoofing.

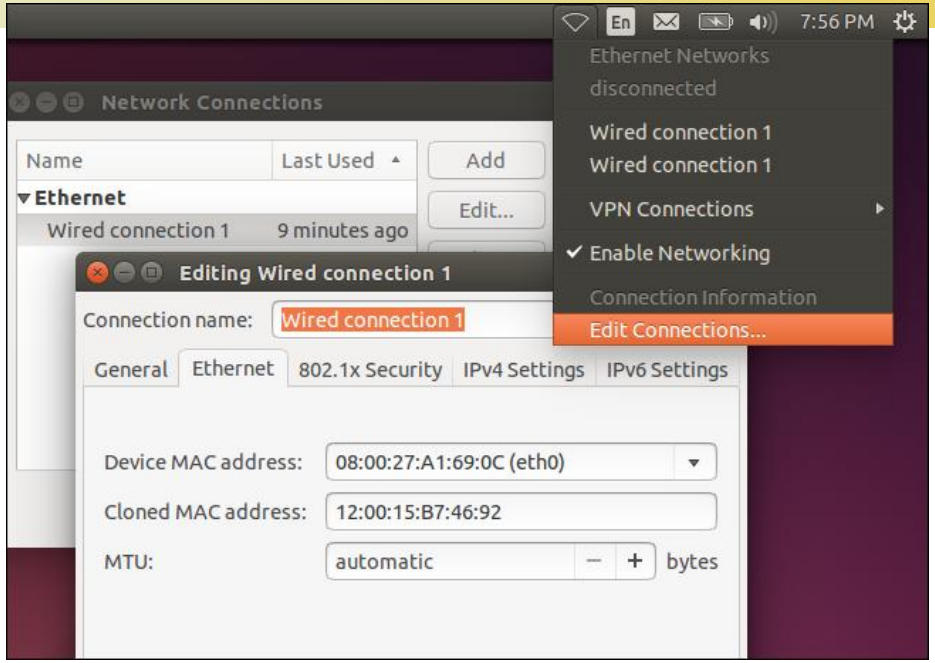
- To prevent third parties from using the MAC address to track devices, Android, Linux, iOS, and Windows have implemented MAC address randomization. In June 2014, Apple announced that future versions of their iOS platform would randomize MAC addresses for all Wi-Fi connections, making it more difficult for internet service providers to track user activities and identities, which resurrected moral and legal arguments surrounding the practice of MAC spoofing among several blogs and newspapers

MAC Spoofing

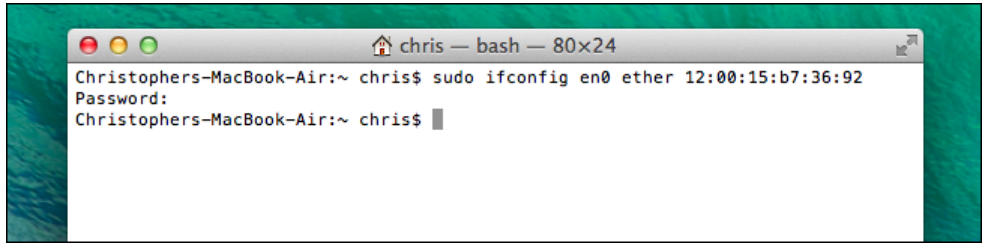
Example



Windows



Linux



MacOS

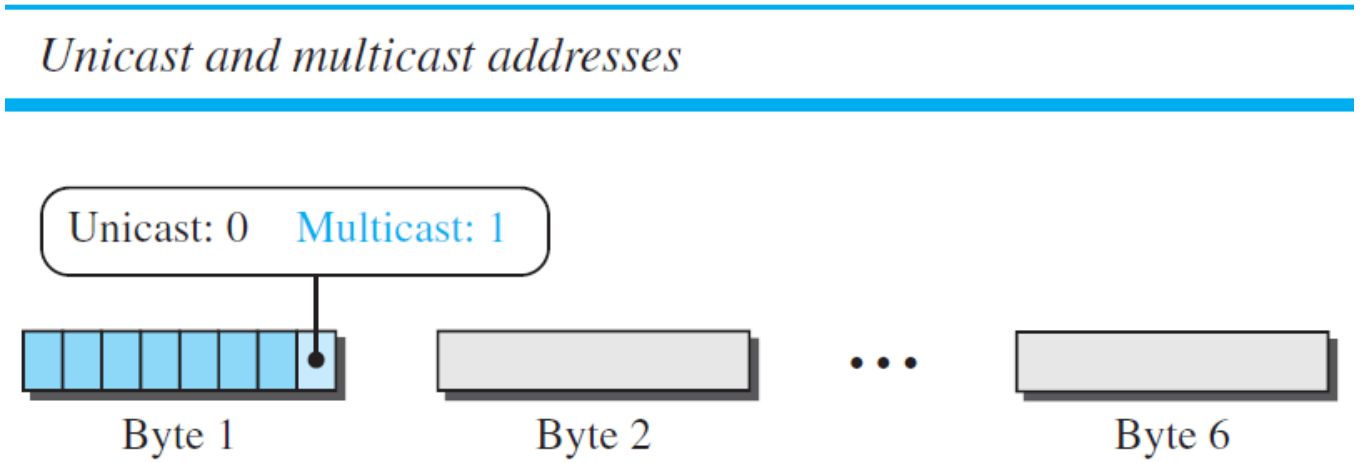
https://www.howtogeek.com/192173/how-and-why-to-change-your-mac-address-on-windows-linux-and-mac/

Using Addresses to Transmit Frames

- The way the addresses are sent out online is different from the way they are written in hexadecimal notation.

- The transmission is left to right, byte by byte;
  - however, for each byte, the least significant bit is sent first and the most significant bit is sent last.
  - This means that the bit that defines an address as unicast or multicast arrives first at the receiver. This helps the receiver to immediately know if the packet is unicast or multicast.

Sent first



MAC Address Types

Example 13.2

Define the type of the following destination addresses:

a. **4A:30:10:21:10:1A**

b. **47:20:1B:2E:08:EE**

c. **FF:FF:FF:FF:FF:FF**

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are Fs, the address is broadcast. Therefore, we have the following:

a. This is a unicast address because A in binary is 1010 (even).

b. This is a multicast address because 7 in binary is 0111 (odd).

c. This is a broadcast address because all digits are Fs in hexadecimal.

Transmission of Addresses (1/2)

Example

- Show how the address 47:20:1B:2E:08:EE is sent out online.

Solution

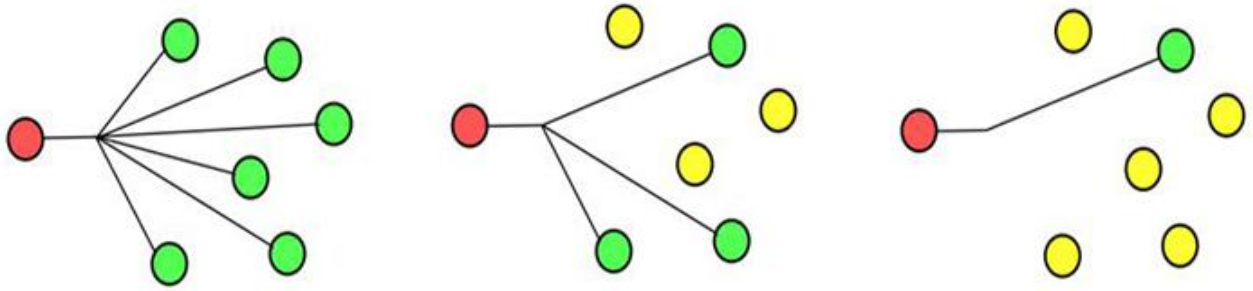
The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

Hexadecimal	47	20	1B	2E	08	EE
Binary	01000111	00100000	00011011	00101110	00001000	11101110
Transmitted ←	11100010	00000100	11011000	01110100	00010000	01110111

Transmission of Addresses (2/2)

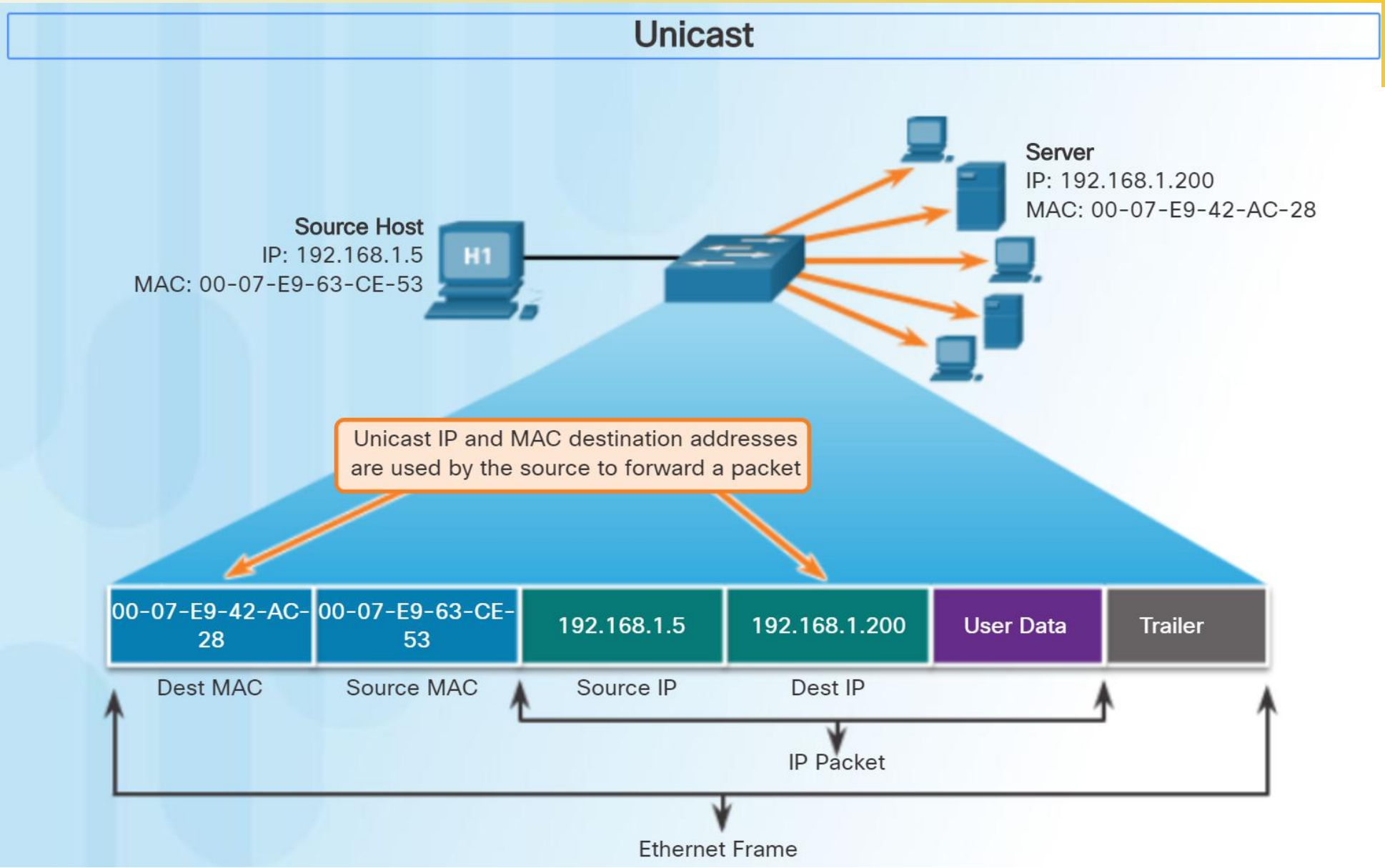
Example

- For Standard Ethernet or traditional Ethernet, the physical layer is shared medium. All transmission will be broadcasted anyway. Then, what do we do about the unicast, multicast and broadcast addresses?
  - In a **unicast transmission**, all stations will receive the frame, the intended recipient keeps and handles the frame; the rest discard it.
  - In a **multicast transmission**, all stations will receive the frame, the stations that are members of the group keep and handle it; the rest discard it.
  - In a **broadcast transmission**, all stations (except the sender) will receive the frame and all stations (except the sender) keep and handle it.

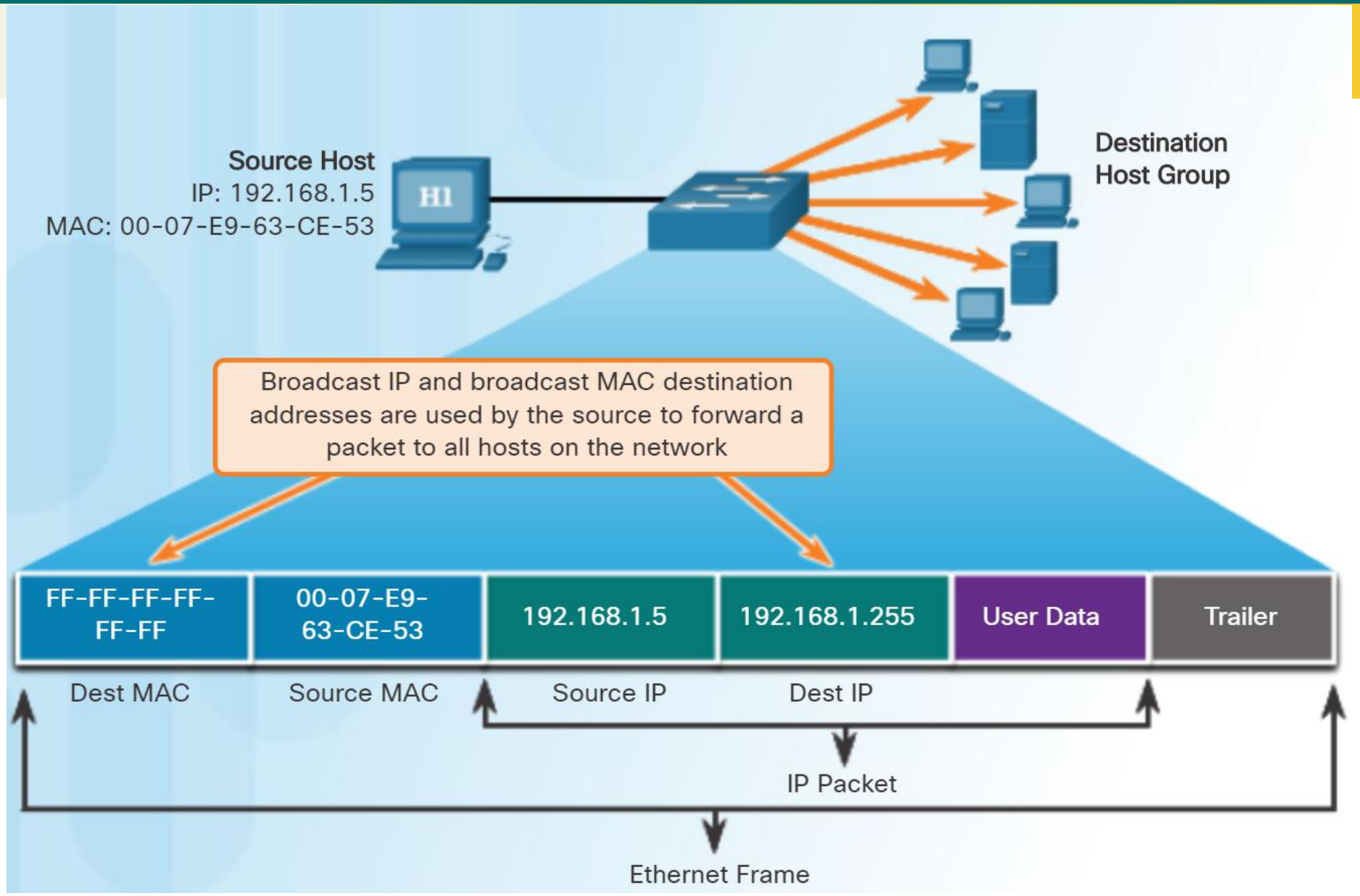


(a) Broadcast (b) Multicast (c) Unicast

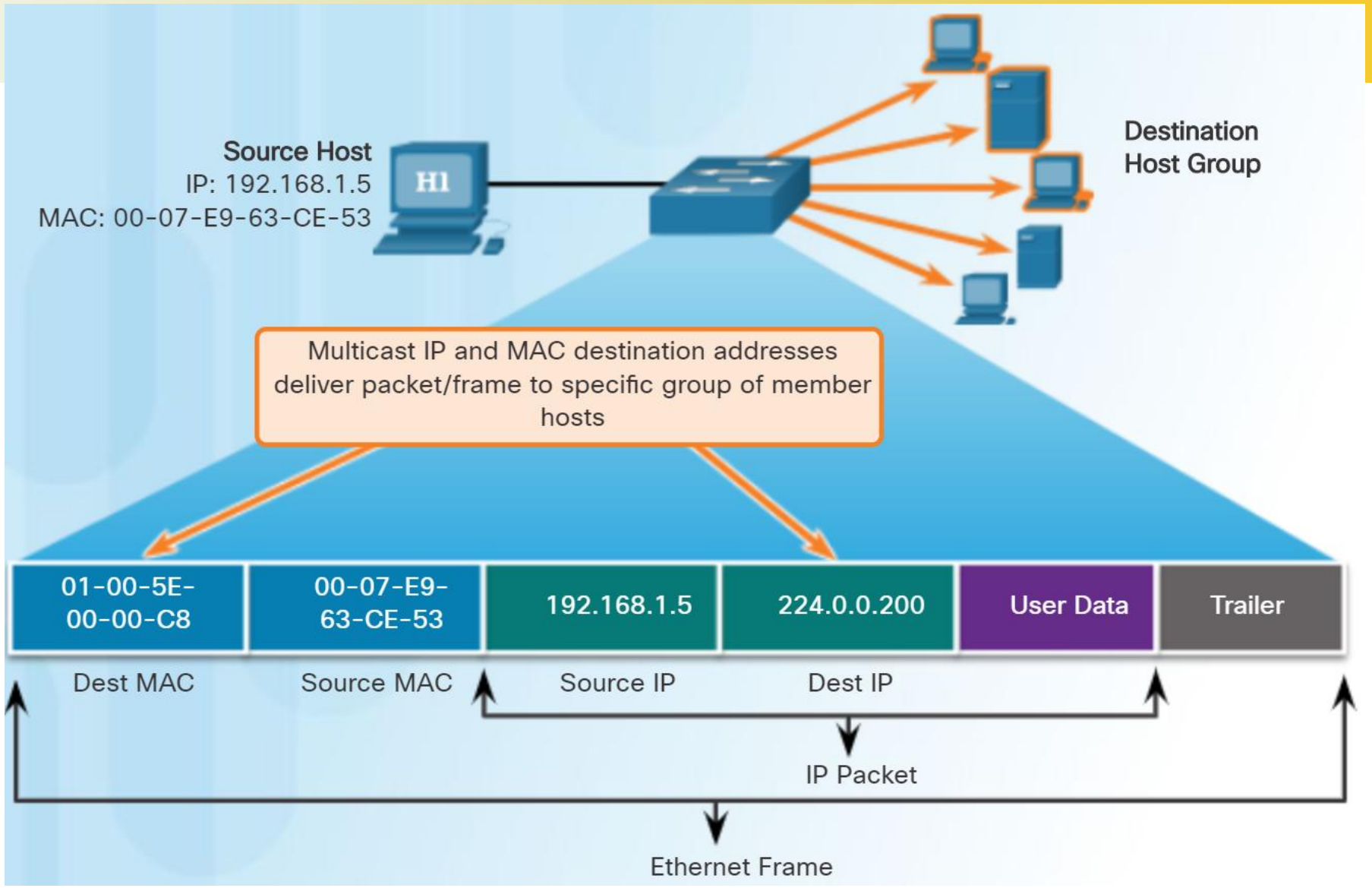
Unicast Frame Transmission



Broadcast Frame Transmission



Multicast Frame Transmission



Broadcast and Multicast

- **Broadcast**
  - Usually broadcast operations involve the use of protocols that requires broadcasting to get information from a network such as DHCP and ARP.
- **Multicast**
  - ✓ The range of IPv4 multicast addresses is 224.0.0.0 to 239.255.255.255. The range of IPv6 multicast addresses begin with FF00::/8. Because multicast addresses represent a group of addresses (sometimes called a host group), they can only be used as the destination of a packet. The source will always be a unicast address.
  - ✓ A host subscribes to a Multicast group using IGMP protocol (not for this course)
  - ✓ Multicast helps the sender by reducing the number of frames that the sender has to send out. Routers/switches will help duplicate multicast frames and send them through ports.
  - ✓ Multicast addresses would be used in multiplayer online gaming, where many players are connected remotely but playing the same game. Another use of multicast addresses is in distance learning through video conferencing, where many students are connected to the same class
  - ✓ The multicast IP address requires a corresponding multicast MAC address to actually deliver frames on a local network.

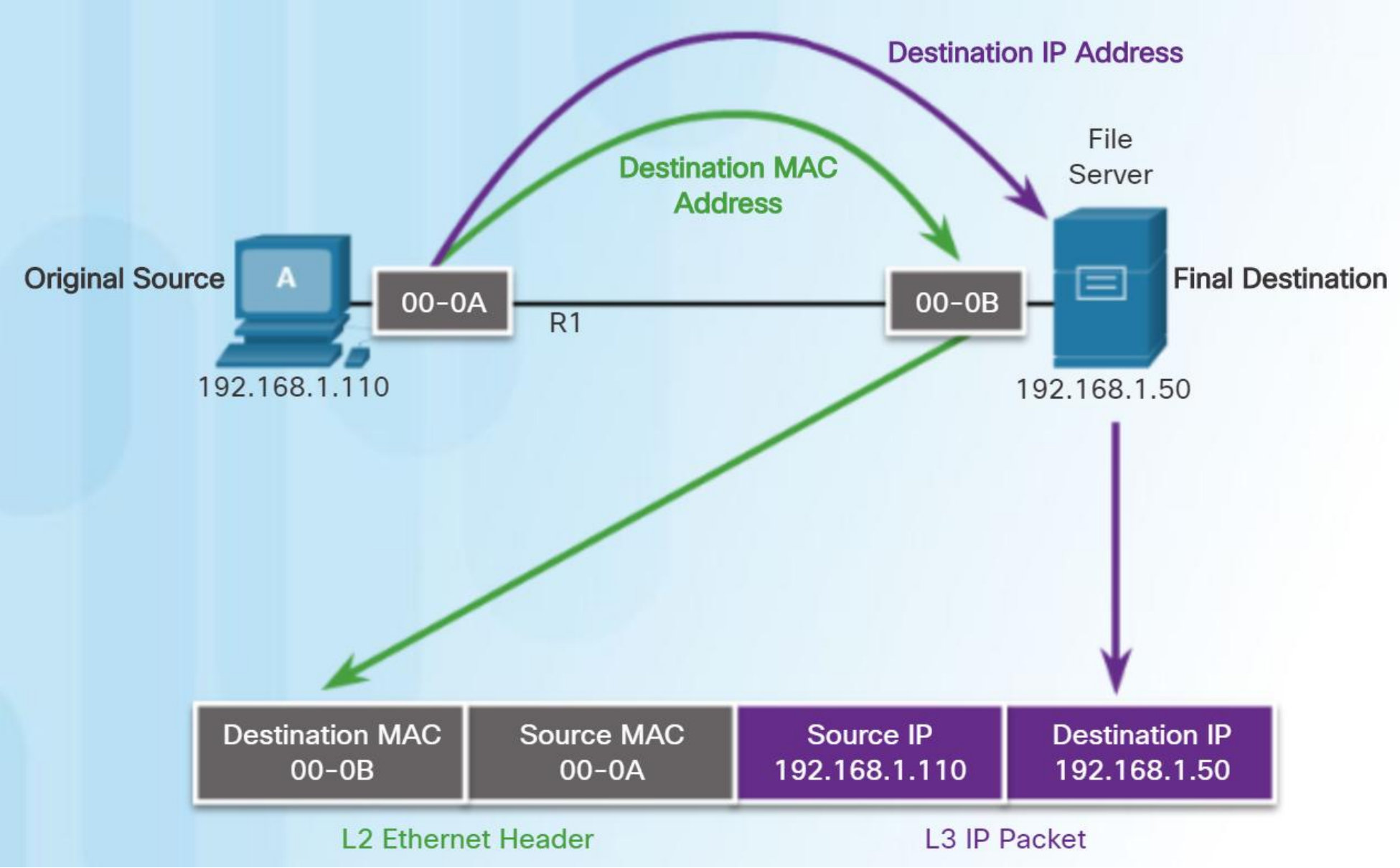
Sending Data from LAN to another LAN

- Sending a frame from one LAN to another LAN (involve OSI Layers)
  - ✓ https://www.youtube.com/watch?v=zhlMLRNY5-4
  - ✓ https://www.youtube.com/watch?v=O7CuFlM4V54
  - ✓ https://www.youtube.com/watch?v=-6Uoku-M6oY

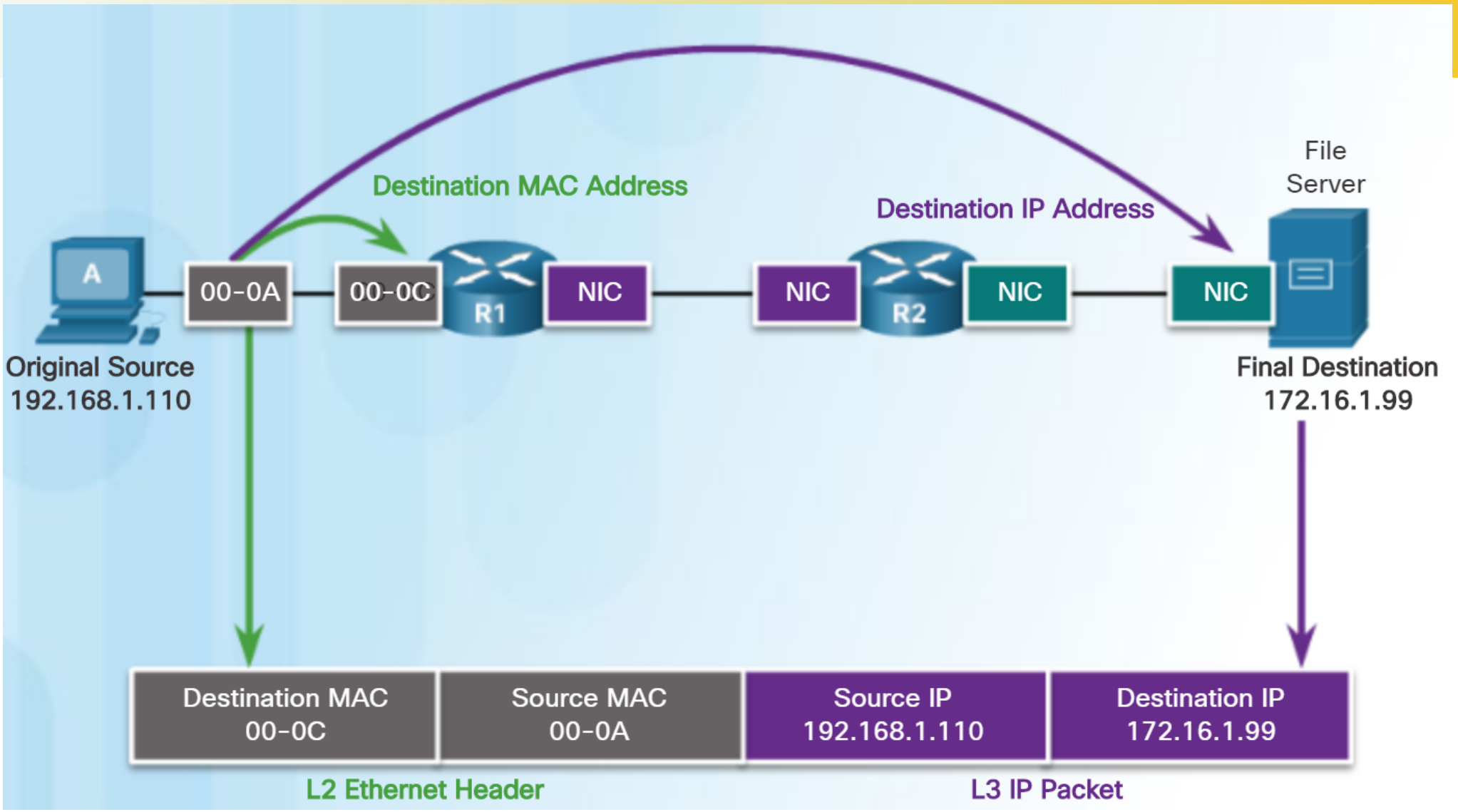
- More info about Ethernet frame structure
  - ✓ https://www.youtube.com/watch?v=_b4dXKB8Pt8
  - ✓ Cisco CCNA1 course

- Data encapsulation
  - ✓ https://www.youtube.com/watch?v=xaKvGnnuYmk

Communicating on a Local Network



Communicating on a Remote Network



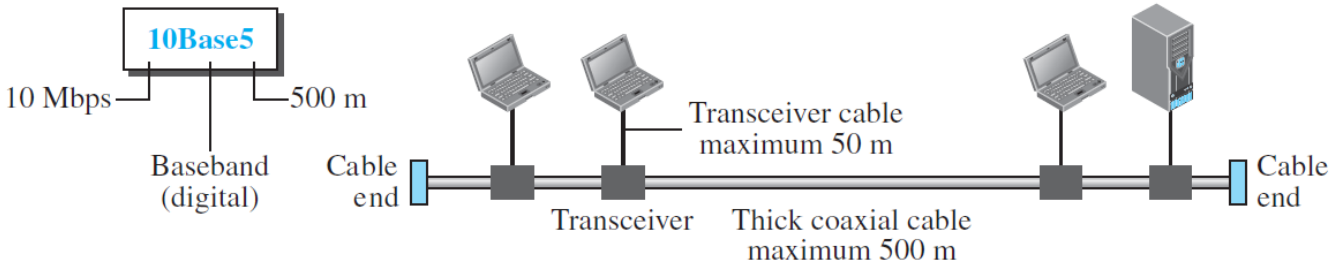
Standard Ethernet (Access Method)

- Only Standard Ethernet requires CSMA/CD. The Standard Ethernet chose CSMA/CD with 1-persistent method. The sender continues to listen during transmission.
- A station sending a minimum frame length of 64 bytes (512 bits).
  - ✓ **Reason**: If we consider the transmission rate of the Ethernet as 10 Mbps, this means that it takes the station 512/(10 Mbps) = 51.2 μs to send out 512 bits. With the speed of propagation in a cable (2 x 10^8 m/s), the first bit could have gone 10,240 meters (one way) or only 5120 meters (round trip), have collided with a bit from the last station on the cable, and have gone back.
  - ✓ In other words, if a collision were to occur, it should occur by the time the sender has sent out 512 bits (worst case) and the first bit has made a round trip of 5120 meters.
  - ✓ However, in practice, the maximum cable length is 2,500 meters to compensate for the delay in the medium.

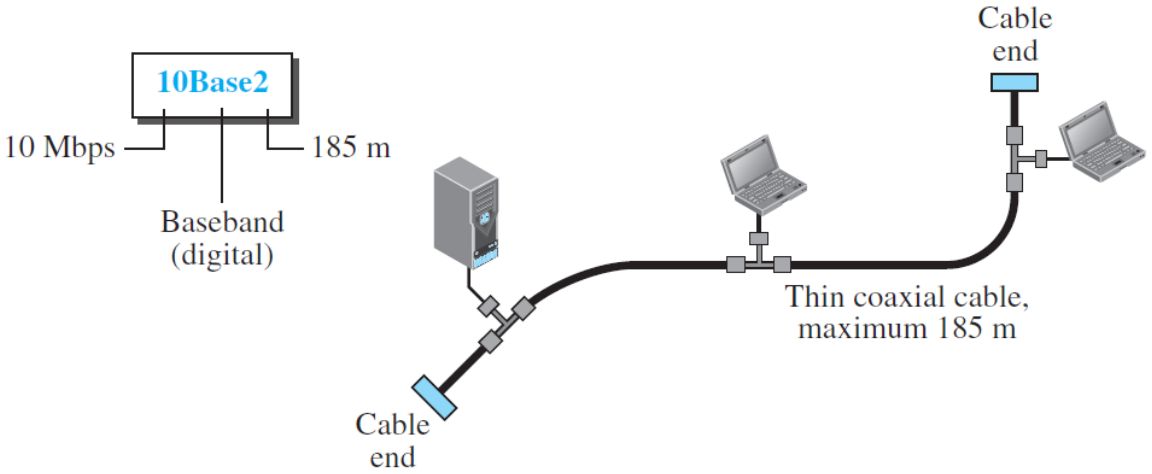
Minimum frame length: 64 bytes	**Minimum data length: 46 bytes**
Maximum frame length: 1518 bytes	**Maximum data length: 1500 bytes**

History (1/2)

- In the 1980s and 1990s several different types of LANs were used. Most of them use the concept of shared medium.



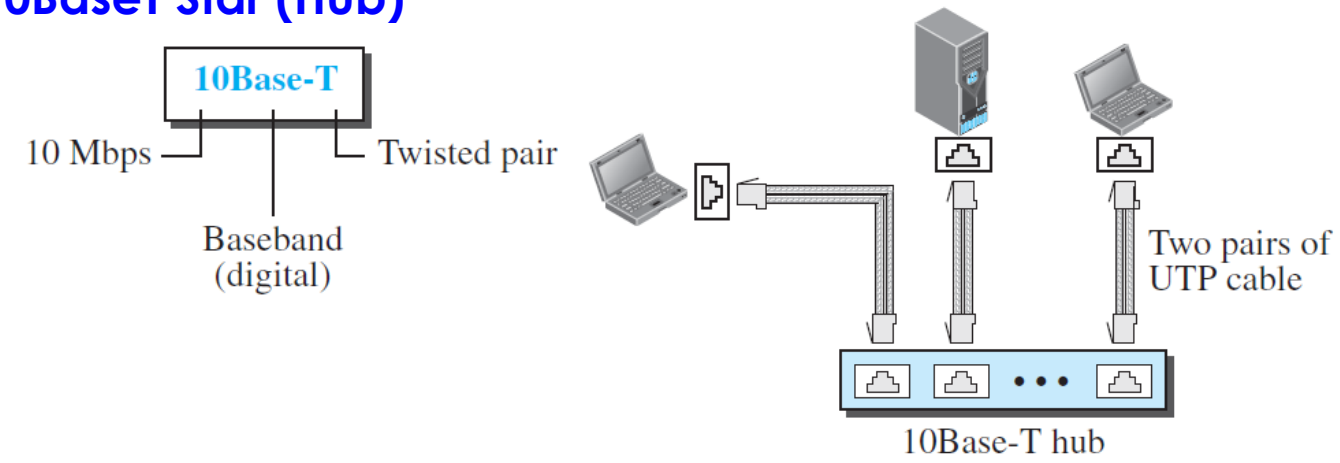
10Base5 Bus (Thick Ethernet)



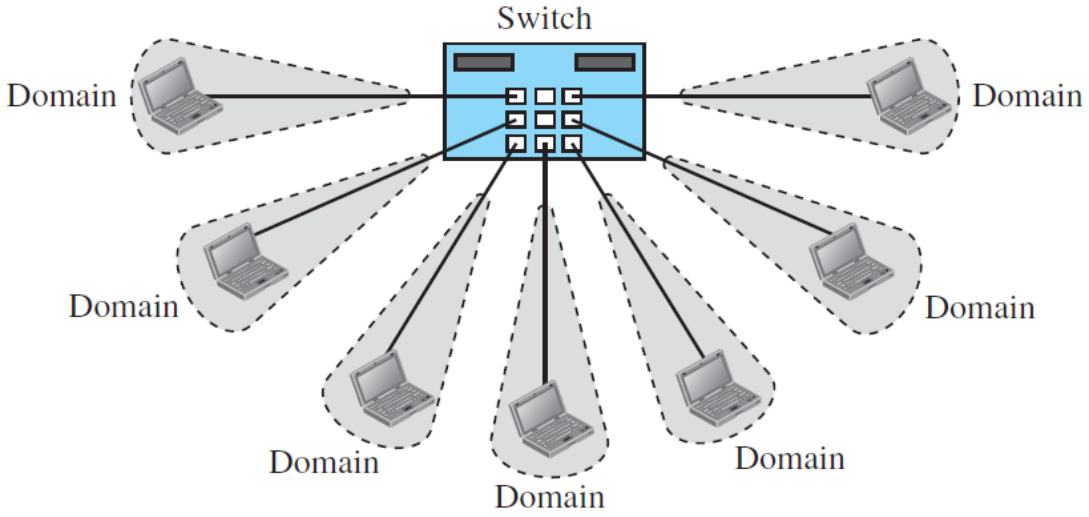
10Base2 Bus (Thin Ethernet)

History (2/2)

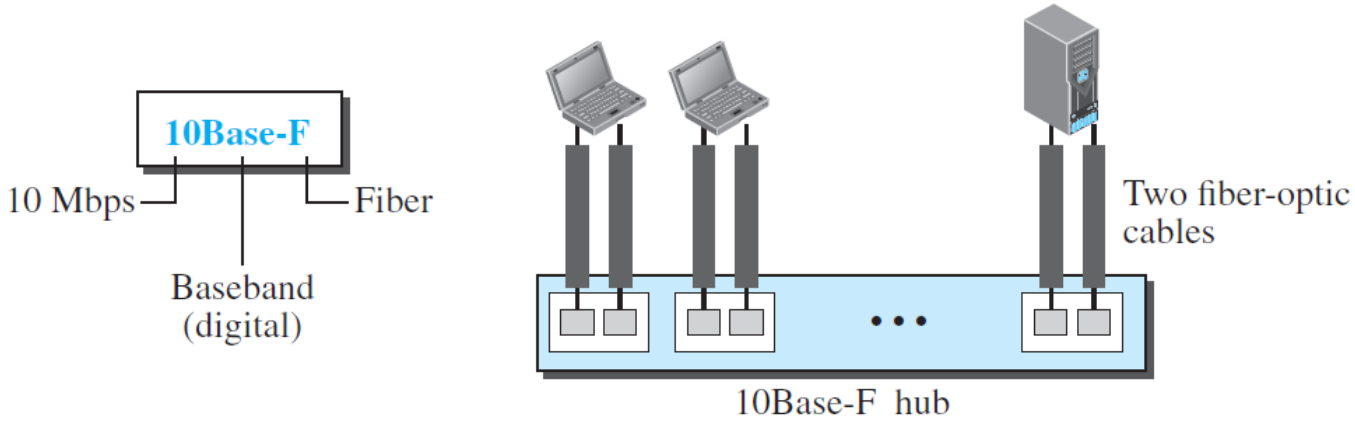
10BaseT Star (Hub)



10BaseT Star (Switch)



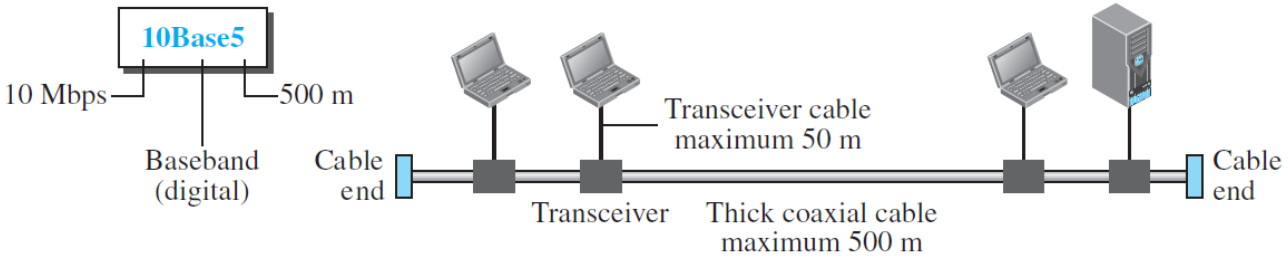
10Base-F Star (Fiber Hub)

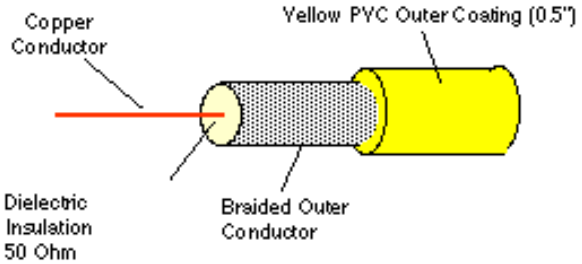


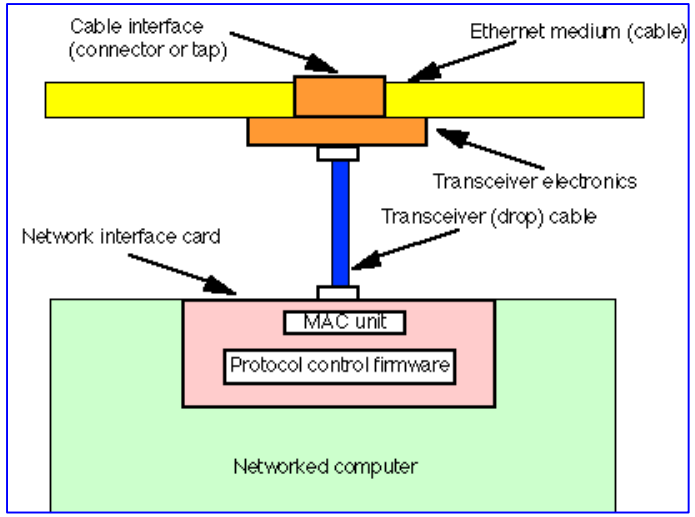
Thick Ethernet (10Base-5)

- The first commercially available Ethernet (1982)

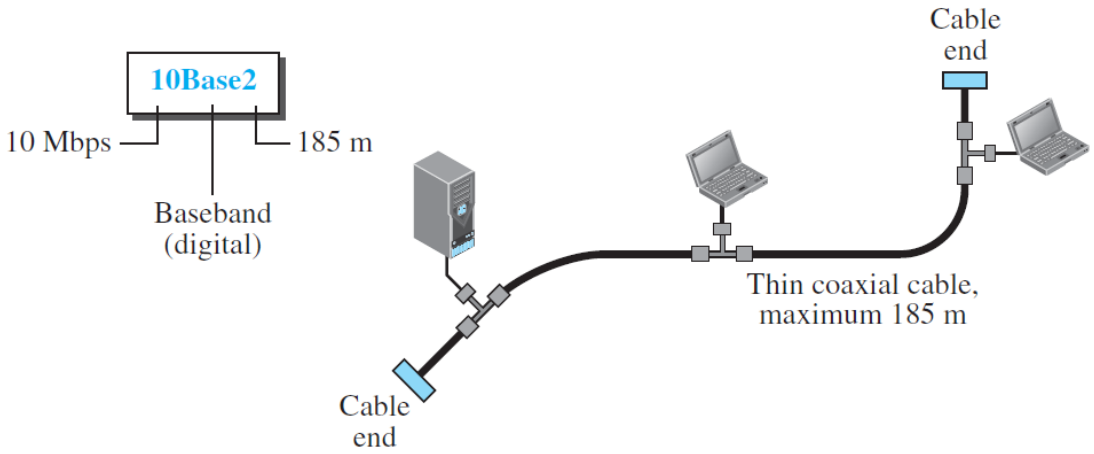
Coaxial cable



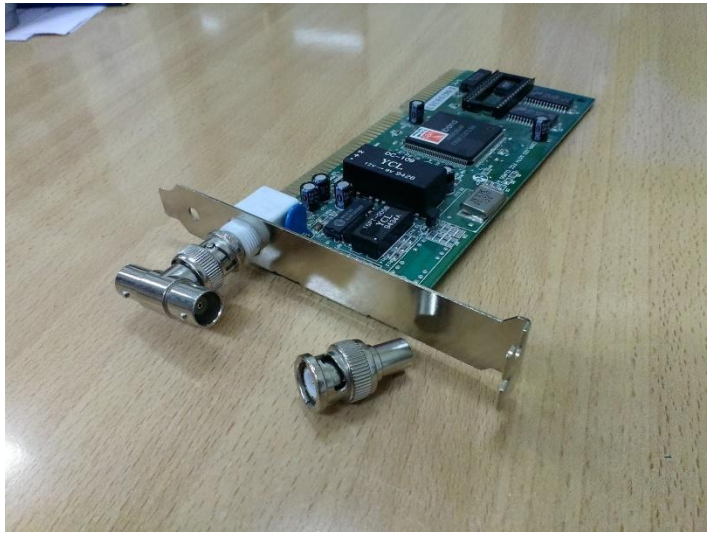




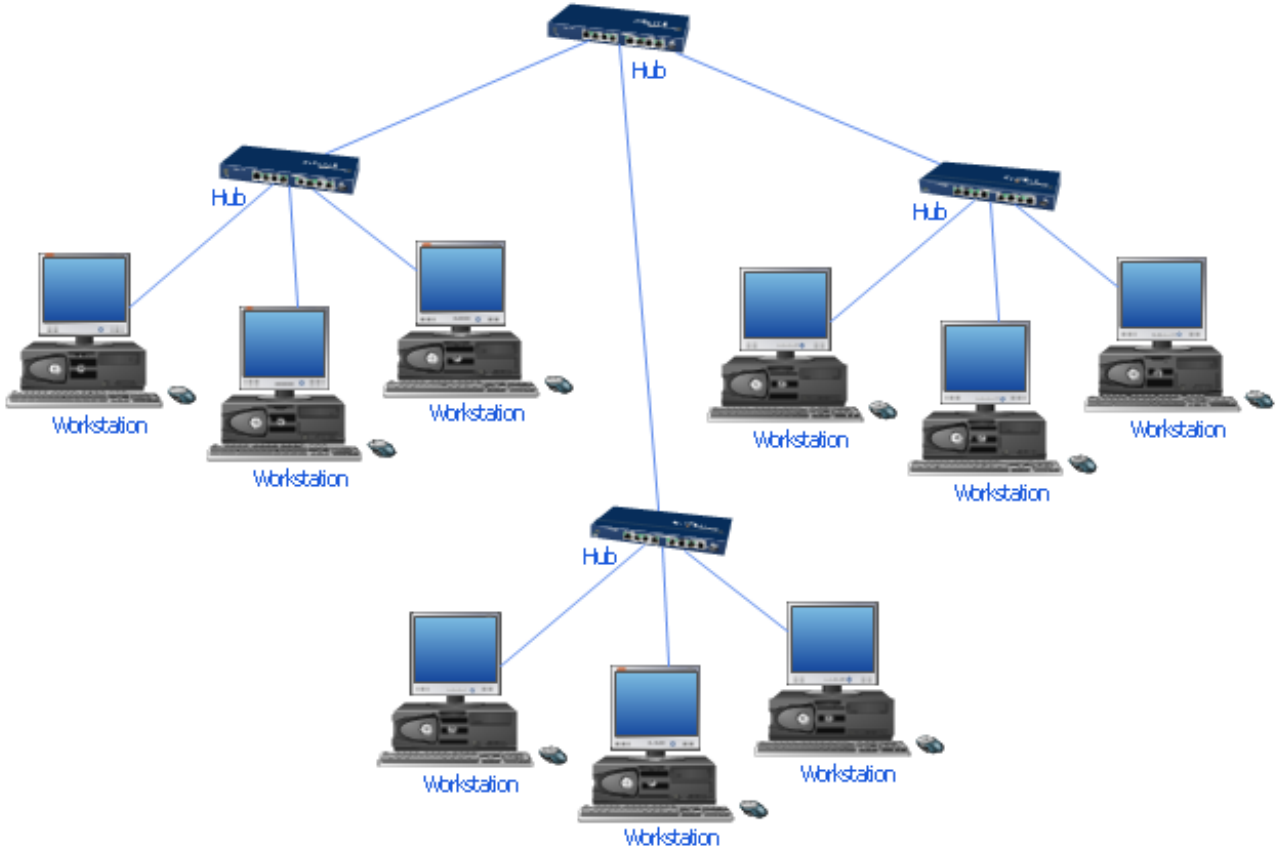
Thin Ethernet (10Base-2)



Coaxial cable

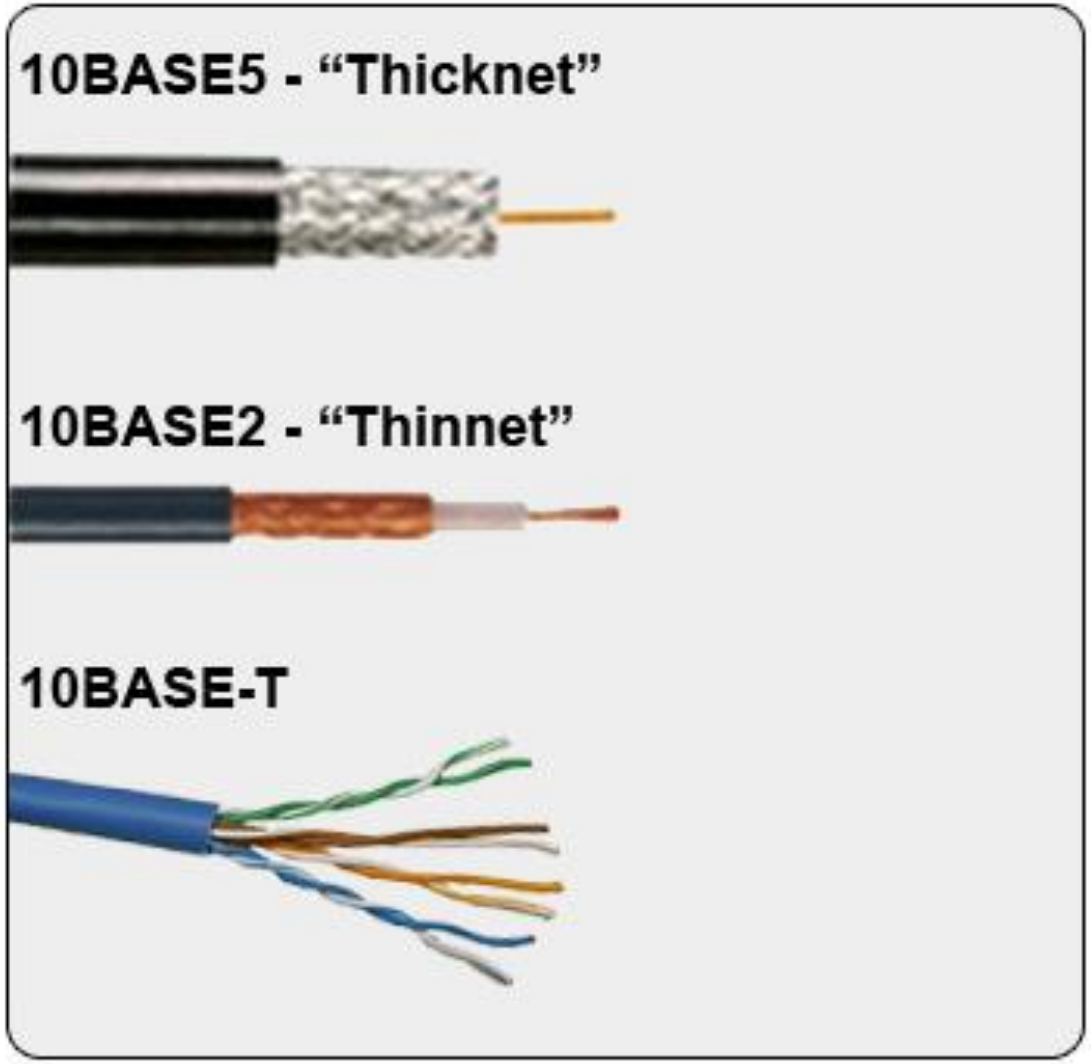


10Base-T Ethernet

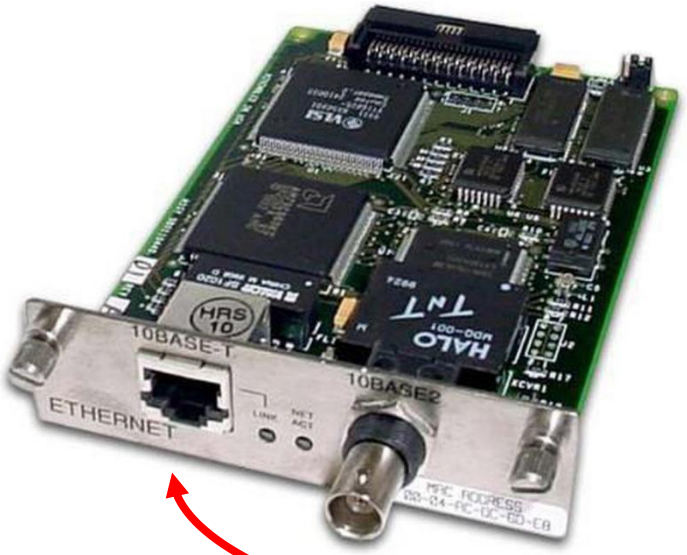


Twisted Pair Cable

Cables



Backward Compatiability

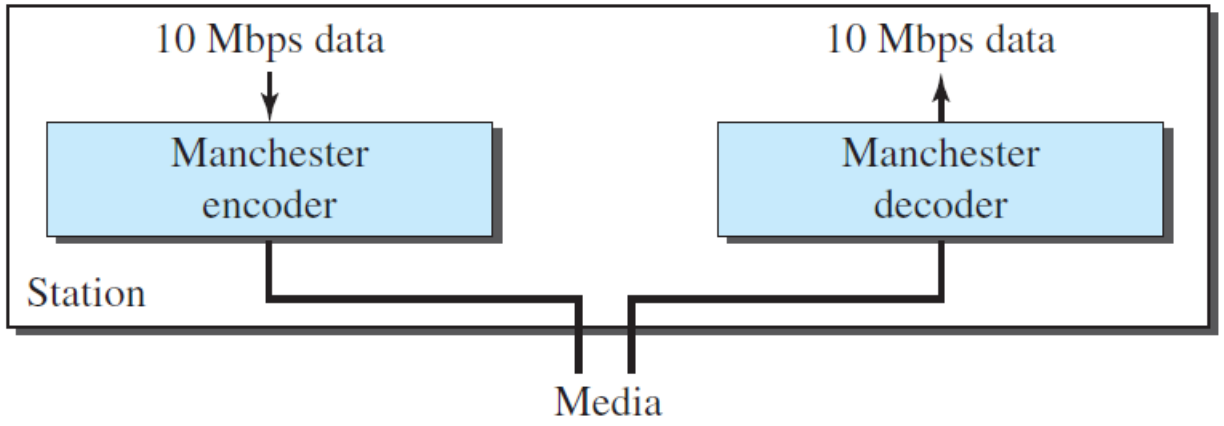


Standard Ethernet

Physical Layer

Table 13.1 *Summary of Standard Ethernet implementations*

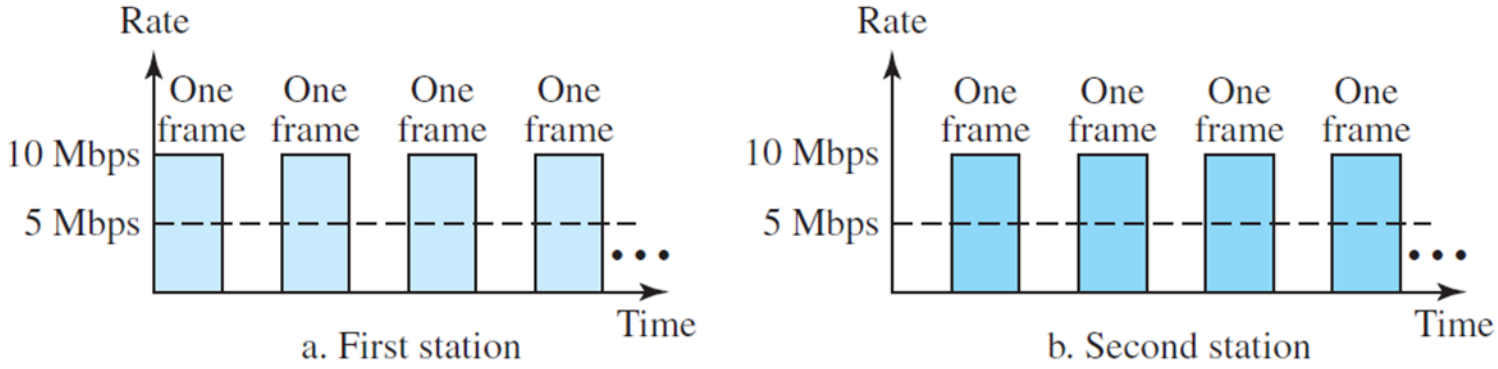
Implementation	*Medium*	*Medium Length*	*Encoding*
10Base5	Thick coax	500 m	Manchester
10Base2	Thin coax	185 m	Manchester
10Base-T	2 UTP	100 m	Manchester
10Base-F	2 Fiber	2000 m	Manchester



Medium Sharing

- **In a Standard Ethernet network**, the total capacity (10 Mbps) is shared among all stations with a frame to send; the stations share the bandwidth of the network. If only one station has frames to send, it benefits from the total capacity (10 Mbps). But if more than one station needs to use the network, the capacity is shared.

- If two stations try to send, each station will get 5 Mbps.

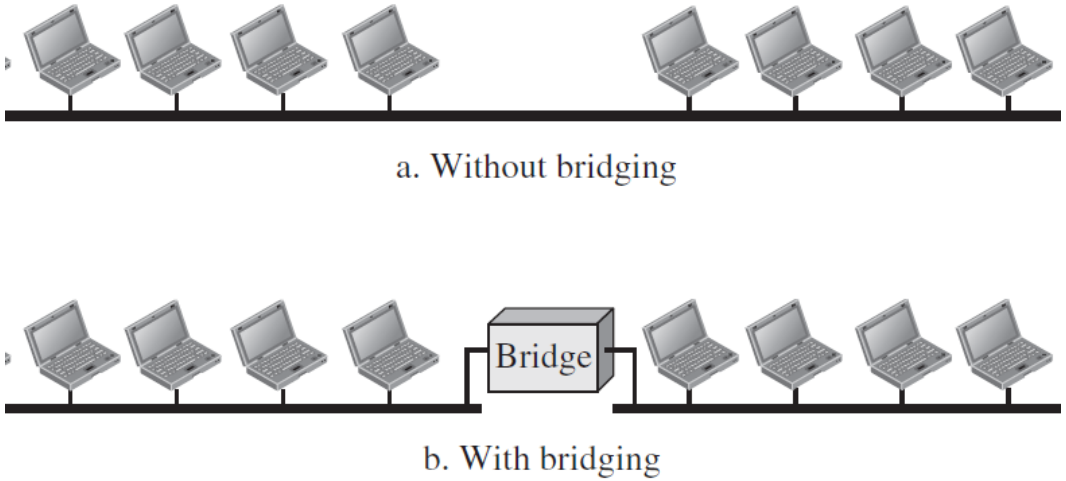


- This means that the more computers you have, the slower you get.

Bridge

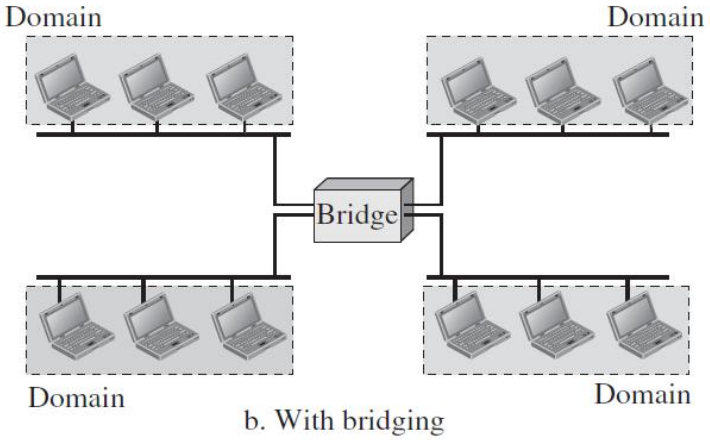
- Placing a bridge to a Standard Ethernet can help separate collision domain. This can increase the data transfer rate within a small segment of the network.

- A **bridge** is essentially a 2 port switch. It can determine whether a frame should be forwarded to the other side of the network.



a. Without bridging

b. With bridging



b. With bridging

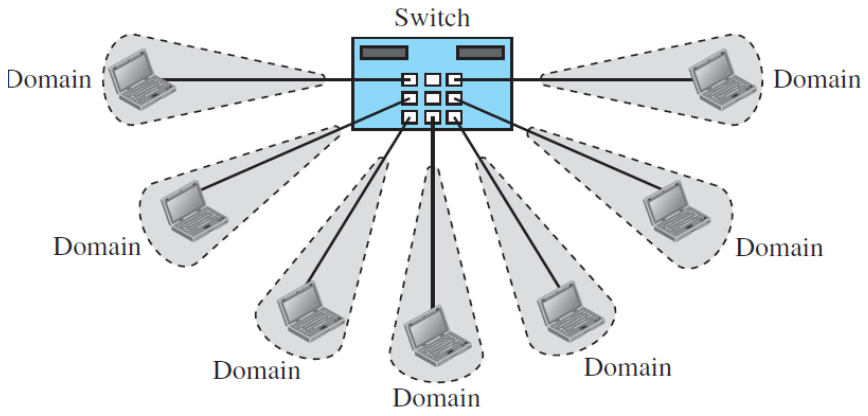
Switched Ethernet

- A switch is essentially a N-port bridge. The collision domain is separated into N subdomains. Each subdomain can enjoy the full 10 Mbps transfer rate.

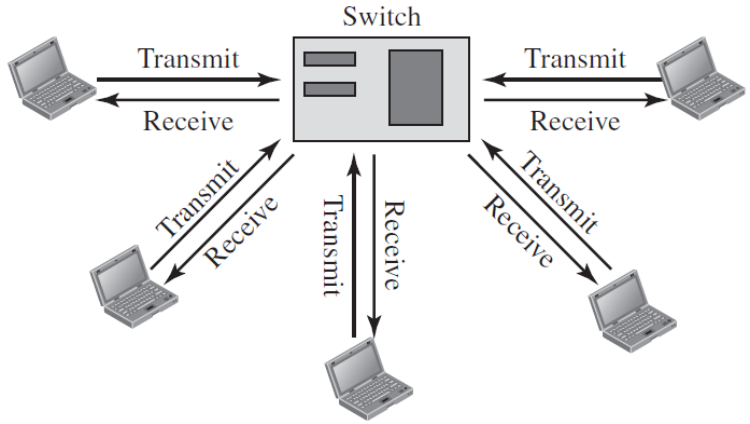
- Collision will never happen. So, CSMA/CD is not needed.

- To fully use the subdomain, full duplex transmission is implemented.

Switched Ethernet



Full-duplex switched Ethernet

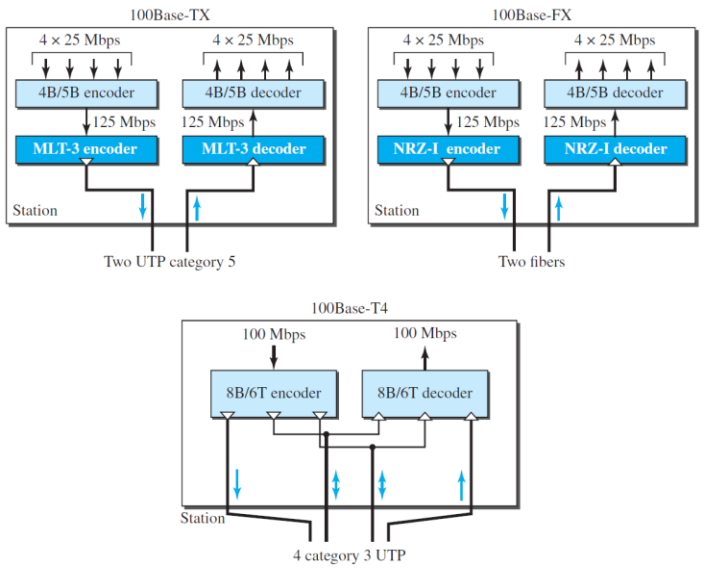


Fast Ethernet (100 Mbps)

- Hub-based Fast Ethernet
  - The minimum frame size is still 64 bytes. However, the frames will be sent **10 times faster.**
  - So, with the influence of CSMA/CD, the maximum length of the cable is reduced from 2,500 meters to 250 meters. In practice, we use the limit of 100 m.
- Switched Fast Ethernet
  - CSMA/CD is not needed.
  - The link-layer switch receives a frame from a source host and stores it in the buffer (queue) waiting for processing. It then checks the destination address and sends the frame out of the corresponding interface.
  - Since the connection to the switch is full-duplex, the destination address can even send a frame to another station at the same time that it is receiving a frame. In other words, the shared medium is changed to many point-to-point media, and there is no need for contention.
- Backward Compatibility
  - When a 10 Mbps LAN is bridged (through a switch or hub) with 100 Mbps LAN, Fast Ethernet has autonegotiation function which allows the two devices to negotiate the speed of transmission (usually use the slower one).

Fast Ethernet (Physical Layer)

- How to make the network run faster?
- Answer:
  - Increase the signaling bandwidth (baud rate). However, Manchester encoding needs a 200-Mbaud bandwidth for a data rate of 100 Mbps, which makes it unsuitable for a medium such as twisted-pair cable. So, we replace Manchester encoding with



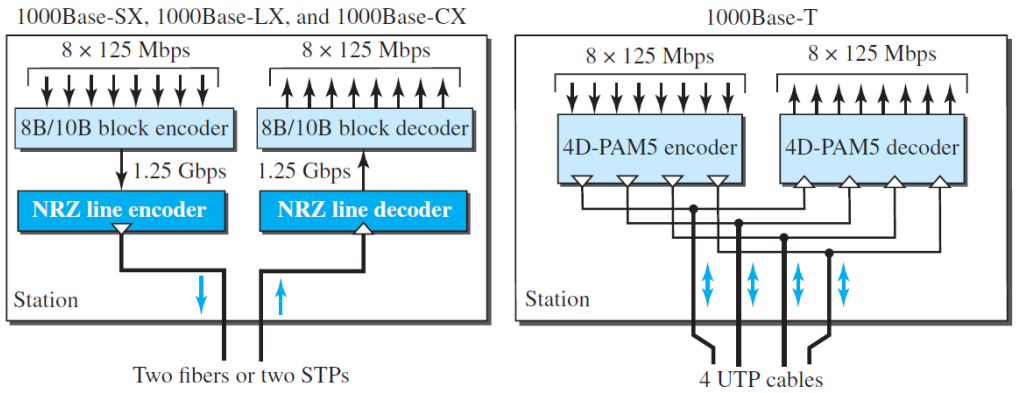
- MLT-3 is used to increase level (increase data rate)
- 4B/5B is used to provide synchronization to support MLT-3.

- 8B/6T is used to provide more levels and synchronization

Implementation	*Medium*	*Medium Length*	*Wires*	*Encoding*
100Base-TX	UTP or STP	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

Physical Layer

- A complete new design of the protocol called IEEE 802.3z.
  - Support speed of 1 Gbps
  - Compatible with Standard and Fast Ethernet using autonegotiation
  - Use the same framing specification.
  - Increase the minimum frame length to 512 bytes (4096 bits) from 64 bytes (512 bits). This is called **Carrier Extension**.
- There is no collision due to the use of switch and full-duplex mode. There is no Gigabit Ethernet hub.
- If the frame length is not 512 bytes, **padding** will be applied. However, multiple frames can be joined together to form a larger frame. This is called **Frame Bursting**.



Implementation	*Medium*	*Medium Length*	*Wires*	*Encoding*
1000Base-SX	Fiber S-W	550 m	2	8B/10B + NRZ
1000Base-LX	Fiber L-W	5000 m	2	8B/10B + NRZ
1000Base-CX	STP	25 m	2	8B/10B + NRZ
1000Base-T4	UTP	100 m	4	4D-PAM5

10 Gigabit Ethernet

Physical Layer

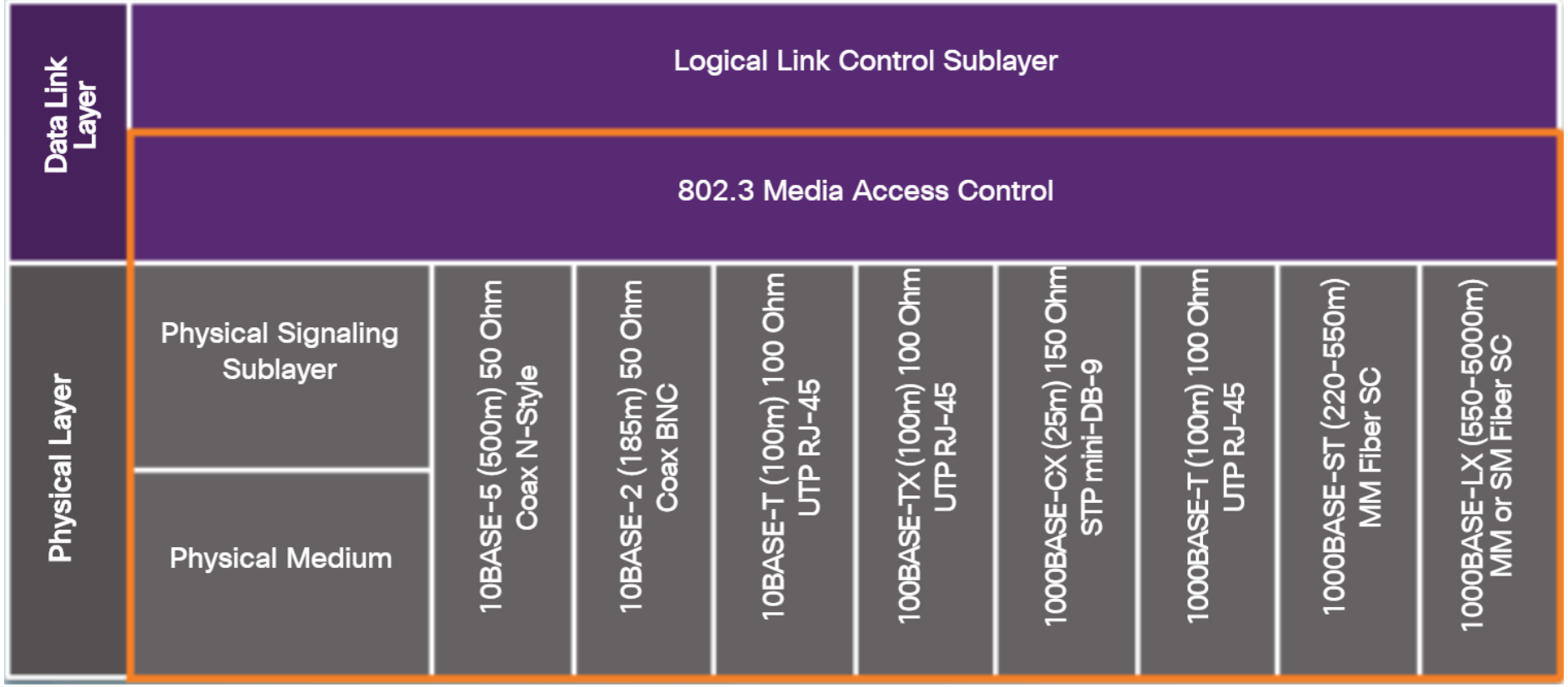
- Only Fiber optics cable can be used. The new standard is called 802.3ae.

- Characteristics
  - Same frame size and format as 1 Gigabit Ethernet.
  - Improve interconnection with MAN and WAN.

Implementation	*Medium*	*Medium Length*	*Number of wires*	*Encoding*
10GBase-SR	Fiber 850 nm	300 m	2	64B66B
10GBase-LR	Fiber 1310 nm	10 Km	2	64B66B
10GBase-EW	Fiber 1350 nm	40 Km	2	SONET
10GBase-X4	Fiber 1310 nm	300 m to 10 Km	2	8B10B

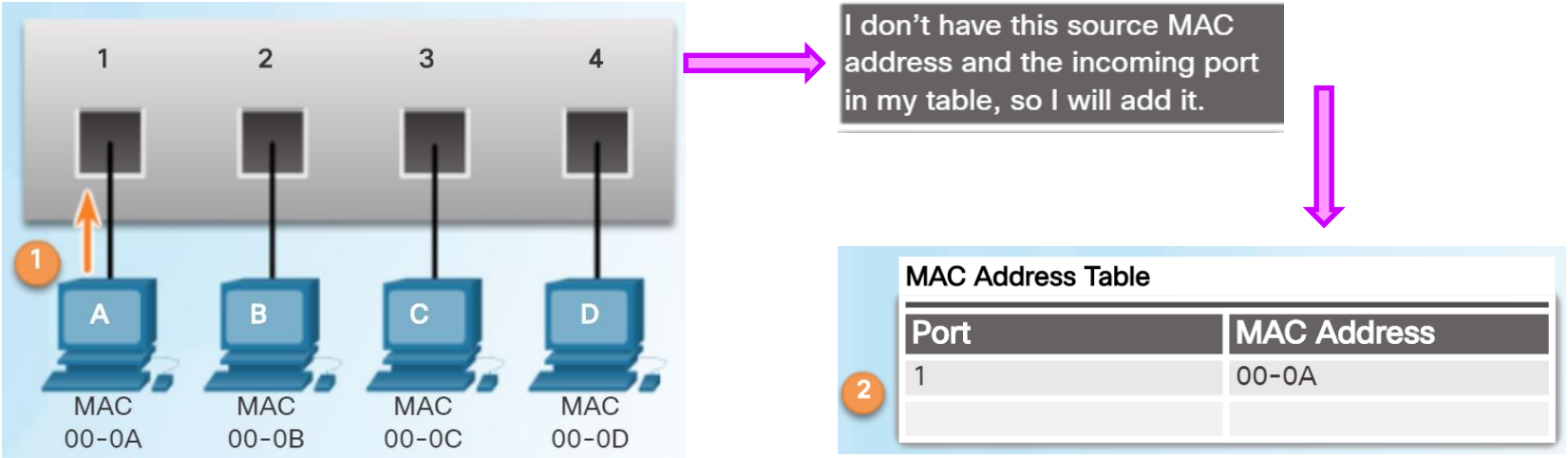
Summary of Standards

Physical Layer



Ethernet Switch

- **Hub and Switch operation**
  - https://www.youtube.com/watch?v=eMamgWllRFY
- **Learning MAC address**
  - The switch dynamically builds the MAC address table by examining the source MAC address of the frames received on a port. If the source MAC address does not exist, it is added to the table along with the incoming port number.
  - Most Ethernet switches keep an entry in the table for 5 minutes.
  - If the source MAC address does exist in the table but on a different port, the switch treats this as a new entry. The entry is replaced using the same MAC address but with the more current port number.

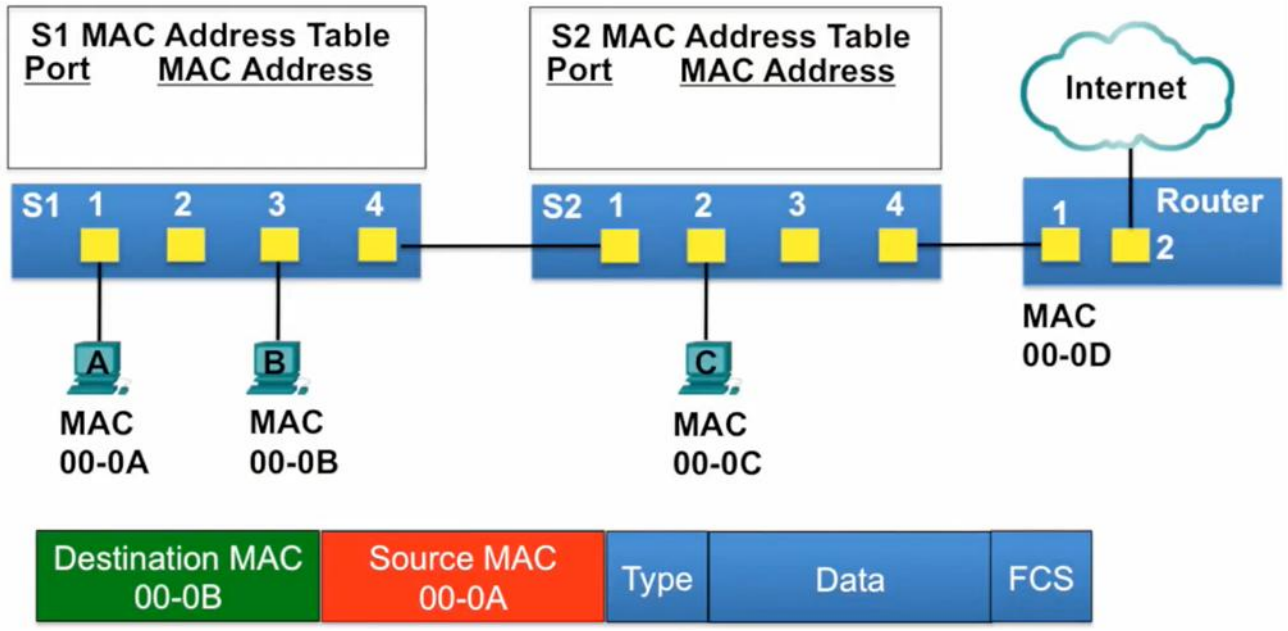


Forwarding

- The switch forwards frames by searching for a match between the destination MAC address in the frame and an entry in the MAC address table.
- If the destination MAC address is in the table, it will forward the frame out the specified port. This is also called frame filtering.
- If the destination MAC address is not in the table, the switch will forward the frame out all ports (connecting to computers) except the incoming port. This is known as an unknown unicast.
- If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

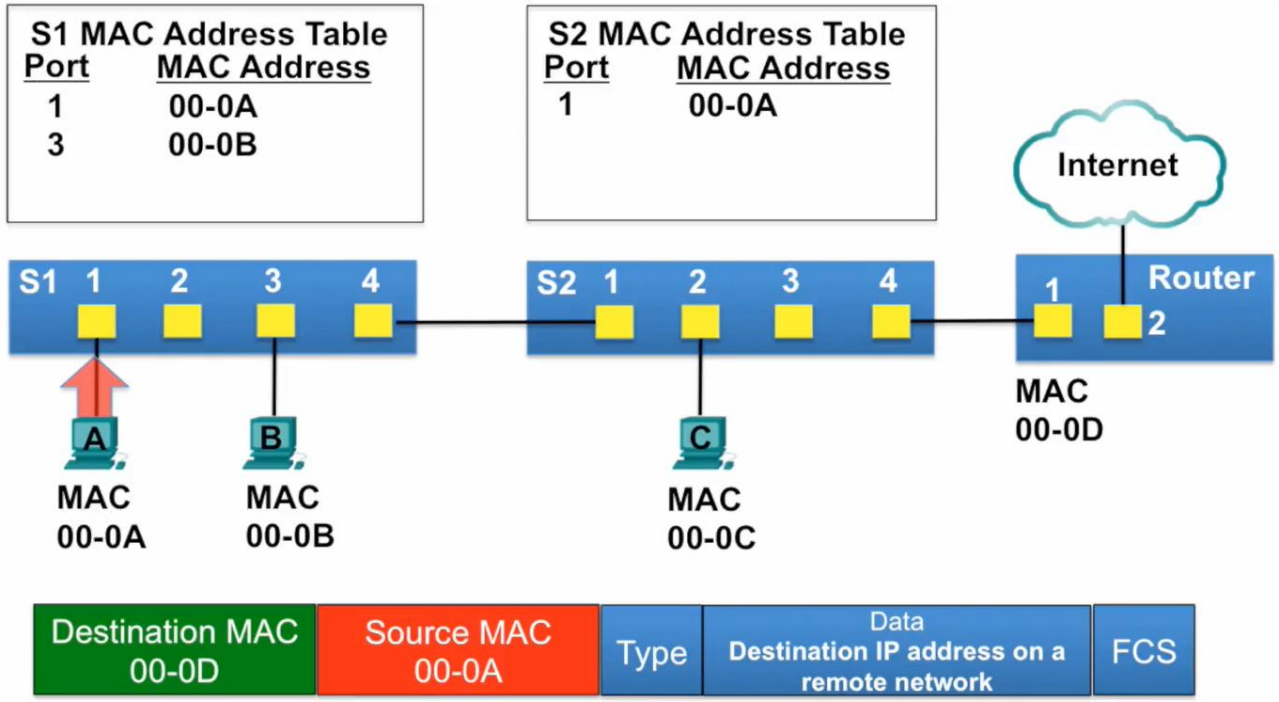
Connected Switches

- A switch can have multiple MAC addresses associated with a single port. This is common when the switch is connected to another switch. The switch will have a separate MAC address table entry for each frame received with a different source MAC address.



Sending Frame outside LAN

- When a device has an IP address that is on a remote network, the Ethernet frame cannot be sent directly to the destination device. Instead, the Ethernet frame is sent to the MAC address of the default gateway, the router.
- The sending device knows about the MAC address of the default gateway through DHCP.



Ethernet Switch

What will a Switch do for each following Scenario?

- Switch receives a frame from a source address not in its MAC table.
  - ✓ Switch puts the source's MAC address in its MAC table along with the associated port.

- Switch receives a frame that has destination address in its MAC table.
  - ✓ Switch forwards the frame to the port associated with the destination address.

- Switch receives a frame that has destination address NOT in its MAC table.
  - ✓ Switch broadcasts the frame to all the ports that is up except the incoming port.

- Switch receives a frame that has destination address is a broadcast address.
  - ✓ Switch broadcasts the frame to all the ports that is up except the incoming port.

Type of Ethernet Switch (1/2)

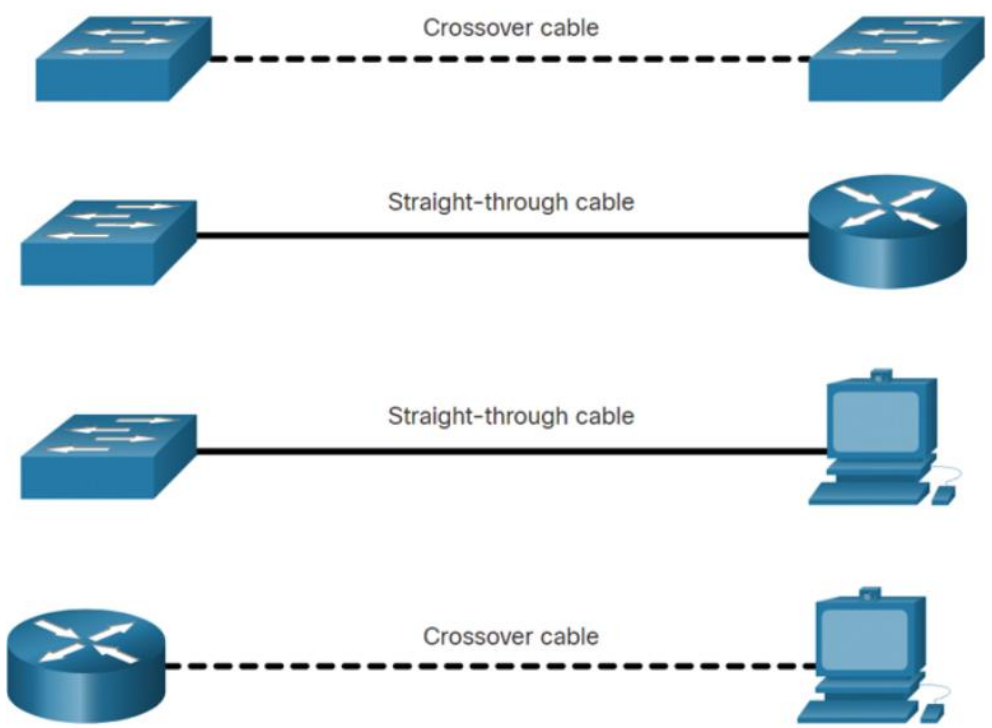
- **Store-and-forward switch**
  - This switch receives the entire frame, checks the frame for error using CRC, and look up the destination address and forward the frame if CRC is valid.
  - Required for Quality of Service (QoS) mechanism to learn the type of frame to provide privilege service.

- **Cut-through switch**
  - This switch forwards the frame before it is entirely received by looking at the destination address of the frame first.
  - Good for fast forwarding that does not need QoS.
  - There are two types
    - Fast forwarding switch (basic type)
    - Fragment-free switch (check first 64 bytes for error – remember CSMA/CD)

Type of Ethernet Switch (2/2)

Description	Store-and-Forward	Cut-Through
Buffers frames until the full frame has been received by the switch.	✓	
Checks the frame for errors before releasing it out of its switch ports if the full frame was not received, the switch discards it.	✓	
No error checking on frames is performed by the switch before releasing the frame out of its ports.		✓
A great method to use to conserve bandwidth on your network.	✓	
The destination Network Interface Card (NIC) discards any incomplete frames using this frame forwarding method.		✓
The faster switching method, but may produce more errors in data integrity therefore, more bandwidth may be consumed.		✓

Auto MDIX



- Most switch devices now support the automatic medium-dependent interface crossover (auto-MDIX) feature.
- When this feature is enabled (by default on switches running Cisco IOS Release 12.2(18)SE or later.), the switch automatically detects the type of cable attached to the port and configures the interfaces accordingly.
- Therefore, you can use either a crossover cable or a straight-through cable for connections to a copper 10/100/1000 port on a switch, regardless of the type of device on the other end of the connection.

Note: Auto MDIX is not Autonegotiation.
Autonegotiation is an optional function on most Ethernet switches and NICs.
It enables two devices to automatically negotiate the best speed and duplex capabilities.